



Indigenous fermented foods of Northeast India: Emerging opportunities in functional foods, microbiota-driven nutrition and safety perspectives

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ARTICLE INFO

Keywords:

Lactic acid bacteria
Ethnic food
Fermented beverages
Fermented bamboo shoot
Fermented soybean
Gut-microbes

ABSTRACT

Traditional fermented foods (TFF) are increasingly recognised for their diverse microbial communities and their contributions to metabolic and gut microbes (GM) health. In Northeast India, spanning eight states and over 150 ethnic groups, these foods are part of daily diets and encompass diverse plant, fish, and meat-based ferments. This review integrates literature from a systematic search up to October 2025, documenting over 154 microbial isolates, primarily lactic acid bacteria, yeasts and spore-formers. These microbes also contribute to nutritional and pharmacological activities. However, the detection of opportunistic pathogens such as *Staphylococcus aureus* and *Clostridium* species highlights concerns not only about microbial contamination but also about broader nutritional and toxicological safety issues arising from non-standardised fermentation practices. This review emphasises the substantial dietary, functional and therapeutic potential of these TFF, while also highlighting the urgent need for mechanistic microbiome studies, standardised fermentation protocols, and clinical validation to ensure safe, evidence-based utilisation.

1. Introduction

Traditional fermented foods (TFF) have emerged as indispensable dietary components due to their significant role in enhancing nutrition, improving food safety, extending shelf life, and supporting metabolic and gut health (Gautam et al., 2025). In the context of an evolving global food environment characterised by escalating lifestyle-related disorders, persistent micronutrient deficiencies, progressive erosion of culinary diversity and increasing reliance on commercially processed food systems, TFF have gained renewed scientific and public health attention. Their relevance is particularly pronounced in India, especially in the North-East (NE) region, where food traditions are intimately linked to ecological richness and ancestral knowledge systems (Bordoloi et al., 2025). NE India spans the Indo-Burma and Himalayan biodiversity hotspots and comprises eight states: Assam, Arunachal Pradesh, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim, and Tripura. As shown in Fig. 1, the region exhibits a rich diversity of traditional fermented foods, with variations in their distribution across different states, reflecting the influence of diverse climatic conditions, biological resources, and indigenous culinary traditions maintained by more than 150 ethnic

communities. Unlike mass-produced fermented products, TFF from this region, made from substrates such as soybeans, bamboo shoots, locally grown vegetables, fish, meat, and native grains, is produced through intricate, community-guided processes shaped by natural microbial reservoirs. This biocultural fermentation system aligns with a One Health perspective, wherein environmental microbiomes, human health, and food systems are intrinsically linked (Banerjee & Van der Heijden, 2023). During traditional fermentation, microorganisms originating from soil, water, plant surfaces and household environments actively participate, shaping the microbial composition of the final food product (Banerjee & Van der Heijden, 2023). Regular consumption of these microbe-rich fermented foods can modulate the human gut microbiota (GM), thereby linking environmental microbial diversity with host metabolism, immune regulation, and overall physiological health (Paonam & Singh, 2025). As a result, these TFFs are not only valued for their nutritional value but also recognised for their functional benefits, and have long been used as household remedies for conditions such as gastritis, respiratory disorders, arthritis, and diabetes (Paonam & Singh, 2025).

TFFs are produced by the metabolic activities of diverse microbial communities, including bacteria, yeasts, and moulds, collectively known

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<https://doi.org/10.1016/j.fbio.2026.108992>

Received 20 December 2025; Received in revised form 24 April 2026; Accepted 26 April 2026

Available online 27 April 2026

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Abbreviations

AFLP	Amplified Fragment Length Polymorphism; CFU: Colony Forming Units
DGGE	Denaturing Gradient Gel Electrophoresis; FDMs: Fermentation-Derived Metabolites
GC-MS	Gas Chromatography-Mass Spectrometry; GM: Gut Microbiota
LAB	Lactic Acid Bacteria; MEGA: Molecular Evolutionary Genetics Analysis; NE: North-East; PCR: Polymerase Chain Reaction; qPCR: Quantitative Polymerase Chain Reaction
RAPD	Random Amplified Polymorphic DNA; RFLP: Restriction Fragment Length Polymorphism; SCFAs: Short-Chain Fatty Acids; Spp.: Species; TFF: Traditional Fermented Foods

as fermentation-associated microbes (Caffrey et al., 2025). The composition and functional roles of these microbes vary depending on the substrate used, the traditional fermentation techniques employed, and local environmental conditions. During fermentation, these microorganisms convert carbohydrates and other nutrients into a wide range of fermentation-derived metabolites (FDMs), such as lactic acid, acetic acid, and bioactive peptides (Talukdar et al., 2024). These metabolites contribute to improved food safety, enhanced flavor and texture, and increased digestibility, while also promoting greater nutrient bioavailability and antioxidant capacity (Park & Mannaa, 2025). Microbial metabolism further enriches TFFs with essential vitamins, releases free amino acids, and bio-transforms phenolic compounds into structurally modified derivatives with enhanced bioactivity. Short-chain fatty acids

(SCFAs), produced either directly during fermentation or formed through microbial metabolism in the gut following consumption, play a key role in maintaining intestinal barrier integrity, reducing local and systemic inflammation, and regulating metabolic pathways (Patrick et al., 2025).

Additionally, fermentation-associated microbes synthesise bacteriocins and exopolysaccharides that help maintain microbial homeostasis, modulate immune responses, and support overall gut health. Fermentation also promotes the biotransformation of complex polyphenols into more absorbable and biologically active forms, thereby enhancing the therapeutic potential of fermented foods (Patrick et al., 2025). These foods influence the GM by delivering live microorganisms, microbial metabolites, and fermentation-derived bioactive compounds that interact with host signalling pathways involved in immunity, metabolism, and intestinal barrier integrity (Mukherjee et al., 2024). Such interactions regulate key molecular pathways, including NF- κ B, AMPK, and SCFA-mediated signalling, linking fermented food intake to modulation of GM composition and host metabolic homeostasis (Tao et al., 2022). The growing understanding of these complex microbe-metabolite-host interactions has been greatly facilitated by recent advances in sequencing-based technologies. High-throughput approaches such as metagenomics, meta-transcriptomics, and metabolomics enable comprehensive analysis of microbial diversity, functional genes, metabolic pathways, and microbial succession during fermentation, including the identification of unculturable microorganisms and bioactive metabolites. These tools help associate specific microbes with probiotic traits and metabolite production. However, limitations such as sequencing bias, incomplete functional annotation, and reliance on predictive analysis remain, highlighting the need to complement omics-based findings with culture-dependent studies and experimental validation to ensure accurate interpretation of functional properties.

As in many other regions of the world, the traditional fermented

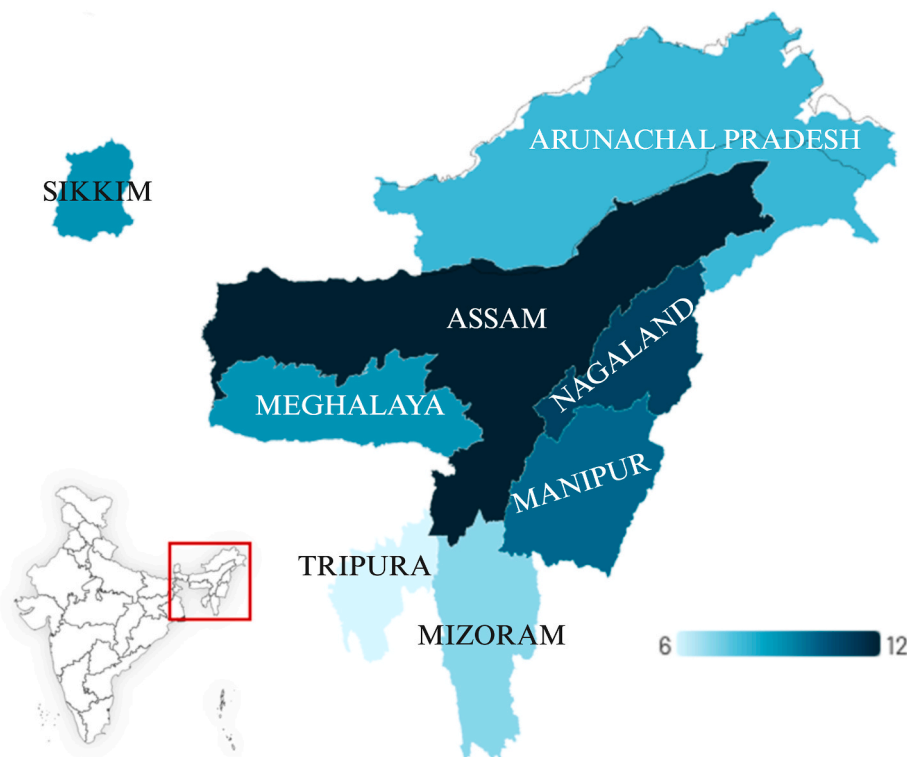


Fig. 1. Map showing the number of traditional fermented foods reported from each state in Northeast India.

The map illustrates the state-wise distribution of documented TFF across Northeast India, including Assam (12), Nagaland (11), Arunachal Pradesh (10), Manipur (9), Sikkim (9), Meghalaya (8), Mizoram (7), and Tripura (6). The color gradient represents the number of reported TFF, ranging from 6 (light shades) to 12 (dark shades), with darker regions indicating higher diversity of traditional fermented foods. The inset map shows the geographical location of the Northeast region within India.

foods of Northeast India have long been associated with digestive health, nutritional enrichment, and community-specific therapeutic beliefs; however, systematic scientific evaluation of these foods remains limited and fragmented. Earlier reviews by Das et al. (2016) and Pohsnem et al. (2023) primarily focused on ethnocultural significance, traditional preparation practices, and general nutritional value, emphasising their socio-cultural importance rather than their mechanistic health relevance. More recent reports, including those by Kabui et al. (2025) and Loying et al. (2024), have provided additional insights into selected fermented products from individual North-Eastern states, particularly regarding indigenous microbial diversity and basic nutritional changes that occur during fermentation. Nevertheless, these investigations do not address microbial ecology at the functional level, metabolite characterisation, or the broader health implications of fermentation-derived bioactive compounds. The present review offers a novel, integrative perspective by systematically compiling available evidence on the nutritional composition, microbial diversity, bioactive metabolites, probiotic and postbiotic potential, pharmacological relevance, and safety of traditional fermented foods from Northeast India. This review emphasises the functional and translational significance of these foods, particularly regarding metabolic health, GM modulation, and immune regulation. In addition, it highlights the current lack of toxicological

validation, standardised analytical approaches, and safety assessment studies, which remain critical barriers to the scientific acceptance and wider utilisation of these traditional products. By bridging ethnobotanical knowledge with modern nutritional, microbiological, and pharmacological evidence, this review provides a comprehensive framework for future mechanistic studies, safety perspectives, and supports the development of scientifically validated, standardized functional foods. Such an approach is expected to facilitate the transformation of traditional fermentation practices into evidence-based nutritional and therapeutic strategies, thereby promoting regional innovation in microbiome-targeted nutrition, functional food development, and food-safety standardization.

2. Search criteria

A comprehensive literature search was conducted across PubMed, Web of Science, SciFinder, and Google Scholar to identify peer-reviewed studies on TFF of NE India and their links to GM and health. Boolean operators were applied using state-specific terms (“fermented foods” AND “Northeast/Assam/Arunachal Pradesh/Manipur/Meghalaya/Mizoram/Nagaland/Tripura/Sikkim”), along with keywords such as “microbial diversity,” “Gut microbiota,” “probiotics,” and “bioactive

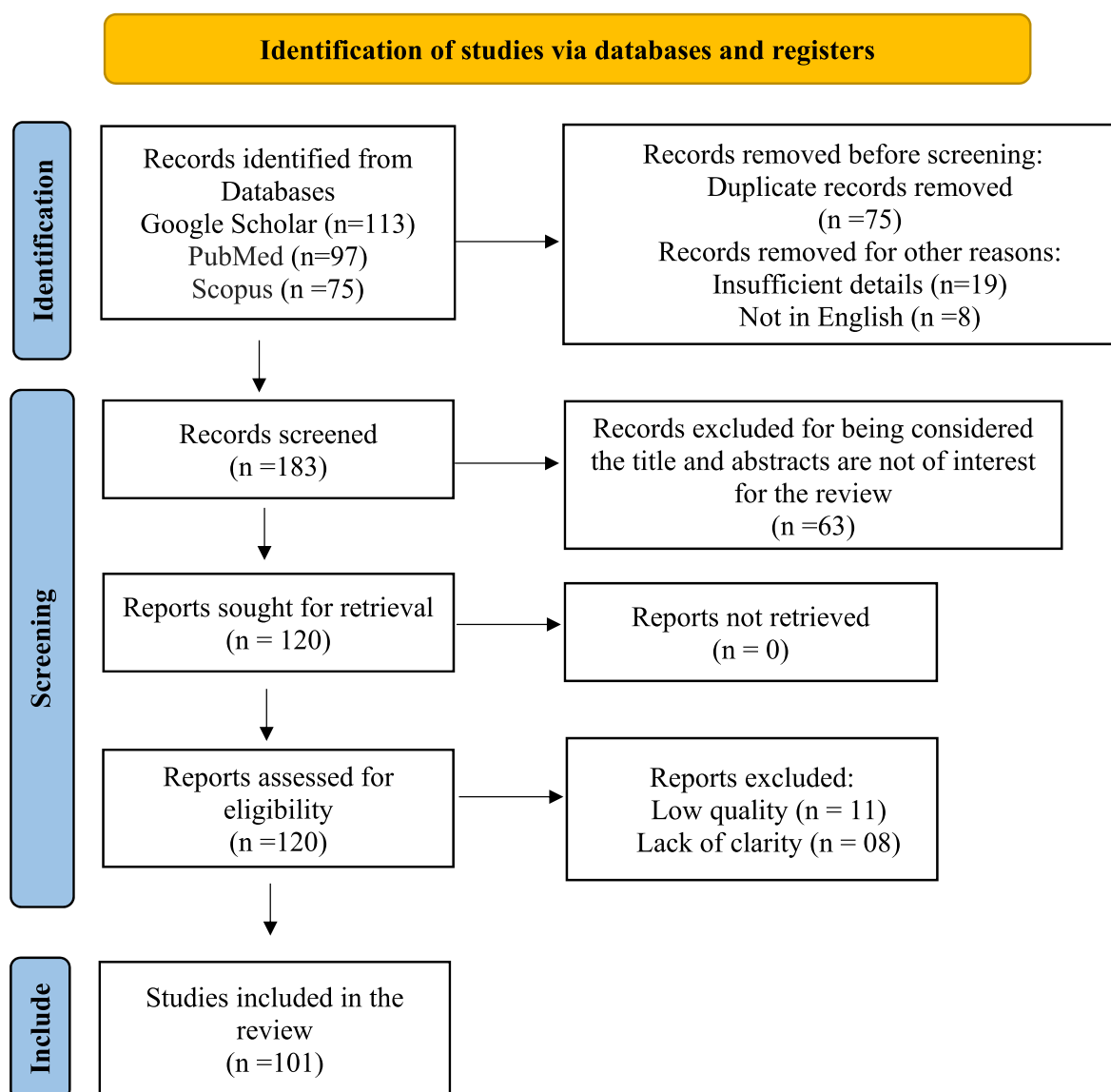


Fig. 2. PRISMA 2020 flow diagram.

metabolites.” Publications up to October 2025 were considered. The process of identification, screening, eligibility assessment, and final selection of studies was conducted according to the PRISMA 2020 guidelines, as illustrated in Fig. 2.

2.1. Inclusion and exclusion criteria

Eligible studies included those reporting microbial composition, FDMs, nutritional properties, or GM-modulating effects of TFF from any of the eight NE Indian states. Only peer-reviewed, English-language original research articles were included. Non-TFF, non-gut-health-related studies, editorials, and duplicates were excluded. Of 285 records retrieved, 183 unique studies remained after duplicate removal, and 101 fulfilled all eligibility criteria for final inclusion.

2.2. Bibliometric analysis

A total of 101 publications meeting the inclusion criteria were subjected to bibliometric analysis. Keyword co-occurrence mapping was performed using VOSviewer (version 1.6.20) to examine the frequency and clustering of key terms, and the corresponding network visualisation is shown in Fig. 3. A minimum keyword occurrence threshold of 5 was applied to include frequently appearing terms. The analysis was conducted using the full-counting method with association-strength normalization, and clustering was performed using the VOS clustering algorithm.

The resulting network visualisation revealed multiple well-defined clusters rather than a single dominant theme, indicating that research on fermented foods in Northeast India is distributed across several interconnected domains. The largest cluster was centred on fermentation, food microbiology, lactic acid bacteria, probiotics, and sequencing-

based approaches, reflecting that the majority of studies focus on microbial characterisation, fermentation processes, and functional properties of fermented foods. The association of this cluster with terms such as metagenomics, metabolomics, and shotgun sequencing suggests an increasing use of advanced molecular and omics-based techniques to understand microbial diversity and metabolic functionality.

A second major cluster was associated with animal, diet, food safety, antimicrobial resistance, and pathogens, indicating that a considerable portion of the literature investigates fermented products derived from animal sources as well as microbiological safety and nutritional quality. The close linkage between these keywords and microbiology-related terms suggests that many studies are oriented toward microbial ecology, preservation, and safety evaluation rather than solely dietary classification. Another distinct cluster included terms such as anti-inflammatory, antioxidants, anti-diabetic, inflammation, and bioactive peptides, demonstrating that a significant body of research examines the functional and pharmacological effects of fermented foods. The frequent co-occurrence of these keywords with terms such as rats, diabetes mellitus, and experimental models indicates that most studies evaluating health benefits are preclinical, highlighting a reliance on animal models to explore metabolic and inflammatory outcomes.

In addition, smaller but clearly identifiable clusters involving probiotics, metabolomics, and gut-related studies indicate an emerging trend toward mechanistic and systems-level investigations, reflecting a shift from traditional descriptive studies toward functional and molecular analyses of fermented foods and their bioactive components. Despite several active research themes, the network showed comparatively weaker representation of keywords related to plant-based fermented foods, milk-based fermentations, and human clinical studies, suggesting that these areas remain underexplored in the current literature. The limited occurrence of clinical and translational terms further

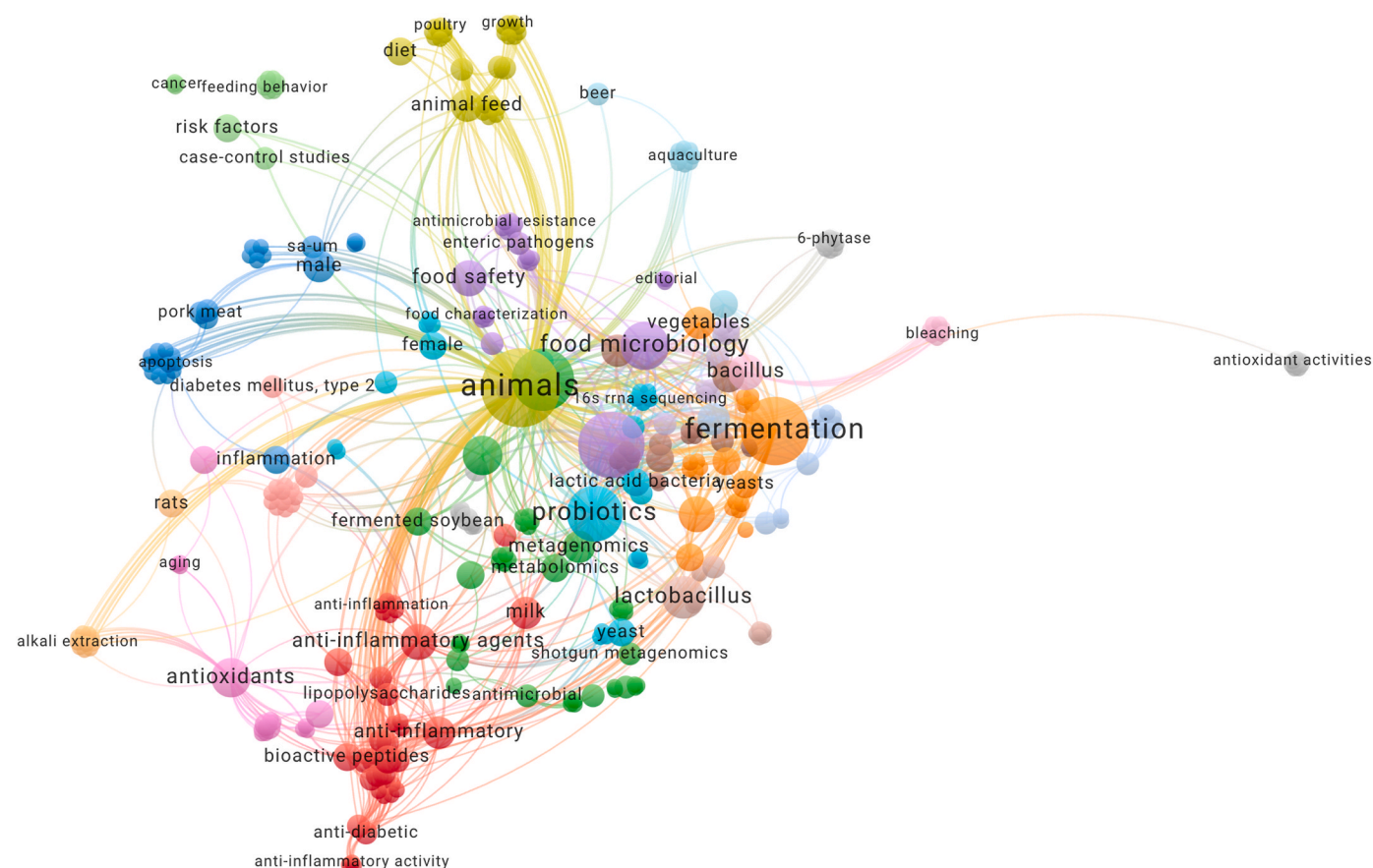
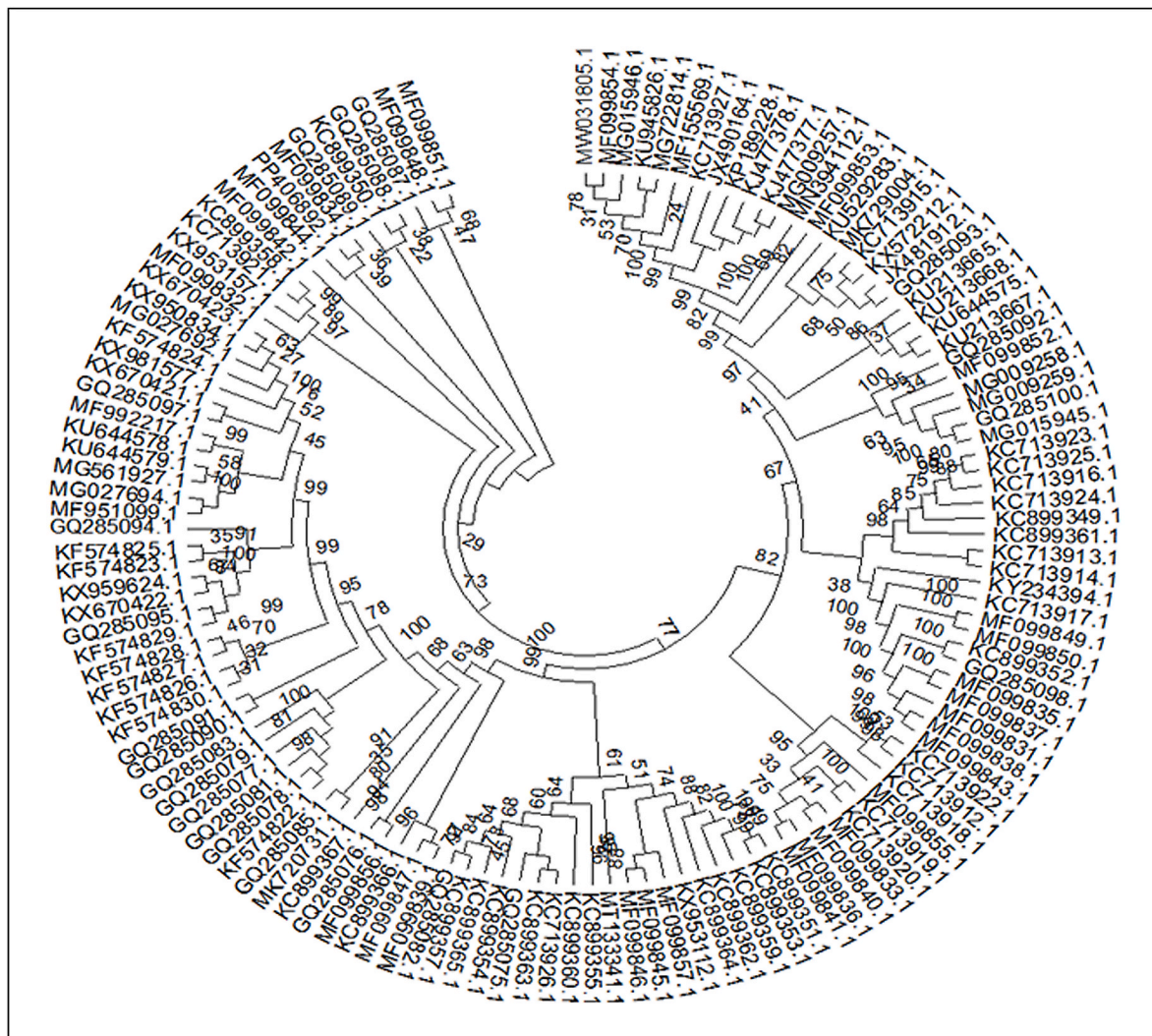


Fig. 3. Cluster network visualisation map from the bibliometric analysis of literature.

3. Microbial diversity of Northeast India traditional fermented foods and their modulatory effects on the gut microbiota

Based on the systematic literature survey and PRISMA-guided study selection, a total of 154 microbial strains, comprising 132 bacterial and 22 yeast isolates, were identified exclusively from the final set of included studies on traditional fermented foods of Northeast India. LAB *spp.*, particularly *Lactiplantibacillus pentosus*, *Limosilactobacillus*

The resulting microbial diversity shows a functionally dynamic ecosystem within fermented foods, particularly dominated by beneficial



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LAB such as *Lactiplantibacillus*, *Limosilactobacillus*, *Lacticaeibacillus*, *Pediococcus*, and *Enterococcus*. These taxa play crucial roles in assuring product quality by synthesising organic acids, bacteriocins, exopolysaccharides, and degradative enzymes that inhibit spoilage microbiota and enhance sensory characteristics (Terpou et al., 2025). Several *Lactobacillus* and *Enterococcus* strains also exhibit probiotic traits, including improved digestive function, immune modulation, and maintenance of intestinal microbial equilibrium (Terpou et al., 2025). In parallel, spore-forming *Bacillus* and *Paenibacillus* spp. contribute robust preservation attributes by producing extracellular enzymes and bioactive metabolites, thereby enhancing the health-promoting potential of fermented-food microbial consortia (Guse et al., 2023).

The mechanistic pathways illustrated in Fig. 5 represent a generalized conceptual framework, derived from previously reported studies on fermented foods and probiotic microorganisms, that describes possible interactions among fermentation-derived microbes, their metabolites, and the GM (Dimidi et al., 2019; Marco et al., 2017). These mechanisms include modulation of gut microbial composition, production of SCFA and other bioactive metabolites, regulation of immune responses, and improvement of intestinal barrier integrity.

Beyond microbial contributions, fermentation significantly enhances the nutritional and functional properties of foods. Complex polysaccharides are enzymatically degraded into oligosaccharides, proteins are hydrolysed into readily absorbable amino acids and bioactive peptides, and lipids undergo modifications that improve digestibility (Balasubramanian et al., 2024). Concurrently, fermentation enriches vitamin content, particularly B7, B11, and B12, and increases antioxidant potential by releasing and structurally altering polyphenolic compounds. Many of these phenolics attain higher bioavailability post-fermentation and selectively favor the growth of beneficial gut microbes while suppressing pathogenic spp. (Terpou et al., 2025). These TFF also function as probiotics, prebiotics, and postbiotics, depending on their microbial and metabolite composition. As probiotics, they deliver live microorganisms that survive gastrointestinal transit, competitively inhibit pathogens, strengthen gut barrier function, and modulate immune responses. As prebiotics, fermentation-derived compounds such as oligosaccharides, peptides, and polyphenols serve as substrates that nourish beneficial gut bacteria, including *Bifidobacterium* and *Faecalibacterium*. As postbiotics, non-viable microbial cells and fermentation by-products, including SCFAs, organic acids, and bacteriocins, exert direct physiological effects, such as anti-inflammatory, antioxidant, and immunomodulatory actions, independent of the presence of live microbes (Ecklu-Mensah et al., 2024). Overall, growing evidence from metagenomic and metabolomic investigations highlights that FDMs act as key signalling mediators within the gut ecosystem, actively modulating host-microbiome interactions and influencing metabolic pathways. These mechanistic insights reinforce the potential of fermented foods as multifunctional dietary strategies that support microbial homeostasis, enhance metabolic resilience, and contribute to long-term health benefits (Debbarma et al., 2025).

3.1. Fermented soybean-based foods

3.1.1. Akhone/axone

The traditional fermented soybean food *Akhone* (*Axone*), widely consumed in Nagaland, exhibits a distinct microbial and metabolite profile influenced by substrate type and native fermentation conditions (Das et al., 2025) (Table 1). A representative image of *Akhone*, illustrating its characteristic appearance and traditional preparation, is shown in Fig. 6. *Akhone* samples collected from seven tribal groups and local markets were analysed to evaluate microbial composition, pH variation, and safety characteristics (Singh et al., 2014). A gradual increase in pH was observed with maturation, with matured *Axone* showing a wider pH range (5.7–8.62) compared with fresh (4.56–7.54) and dried samples (5.58–8.54). Among the different varieties, *Sema* and *Axone* recorded the highest pH (8.02), reflecting variations in traditional

preparation practices and environmental conditions.

Culture-dependent microbial profiling, including total aerobic plate counts, selective culturing, and biochemical identification, revealed a diverse microbial community dominated by *Bacillus* spp. with *B. coagulans* being the most abundant, followed by *B. pentothenicus*, *B. subtilis*, *B. stearothermophilus*, and *B. lentus* (Singh et al., 2014). *Enterococci* were detected in 94.6% of samples, particularly in dried products, with *E. faecalis*, *E. caecorum*, and *E. hirae* as the predominantly *Staphylococci* were present in all samples, mainly *S. xylosus*, *S. epidermidis*, and *S. sciuri*. Antimicrobial profiling of 313 isolates (153 Gram-negative and 160 Gram-positive representing 40 spp.) further confirmed the high microbial diversity associated with *Axone* fermentation (Das et al., 2023).

GC-MS-based metabolomic analysis showed that *Axone* contains prebiotic metabolites and polyunsaturated fatty acids that may influence gut microbial activity and metabolic health (Das et al., 2024). The coexistence of beneficial *Bacillus* spp. and bioactive metabolites suggests that *Axone* may contribute to GM balance while improving nutrient availability through fermentation. Matured *Axone* appeared safer than fresh or dried forms, likely due to prolonged fermentation, which suppresses undesirable microorganisms and stabilises the microbial community.

Despite these functional attributes, safety evaluation revealed several concerns. Nearly one-third of the samples were contaminated with coliforms, including 24 samples positive for *Escherichia coli*. In addition, opportunistic and pathogenic Gram-negative bacteria such as *Aeromonas eucranophila*, *A. hydrophila*, *A. salmonicida*, *Citrobacter* spp., *Enterobacter* spp., *Erwinia ananas*, *Hafnia alvei*, *Klebsiella pneumoniae*, *K. oxytoca*, *Morganella morganii*, *Providencia rettgeri*, *Proteus* spp., *Pseudomonas* spp., and *Salmonella enterica* ser. *Salmonella typhimurium* was isolated (Singh et al., 2014), indicating possible hygiene lapses during preparation or storage and emphasising the need for adequate cooking before consumption. Fungal contamination was detected in 51.8% of samples, with higher yeast and mould counts in fresh and dried products, whereas prolonged fermentation reduced fungal growth and improved safety. Although *enterococci* and *staphylococci* contribute to flavour development, they may also act as opportunistic pathogens or carriers of virulence factors when present in high numbers. Furthermore, 50 out of 313 isolates showed multidrug resistance, suggesting that traditional fermentation products may serve as reservoirs of antimicrobial resistance if processing conditions are not properly controlled (Das, Afzal, et al., 2023). These findings highlight the importance of controlled fermentation, proper storage, and thorough cooking to minimise foodborne risks associated with *Axone*.

3.1.2. Bekang

Bekang, a fermented dry soybean product, is one of the most widely consumed traditional foods of Mizoram, valued for its distinctive taste, nutritional richness, and cultural importance among indigenous communities (Vangchhia et al., 2023). Despite its popularity, scientific studies on its microbial diversity and functional properties are limited. The typical form and physical characteristics of this fermented soybean product are shown in Fig. 6, which demonstrates its dried, compact structure after traditional fermentation. To address this, JDeKa, Mehetre, et al. (2021) analysed *Bekang* samples collected at different fermentation stages (3rd, 5th, and 7th days) using culture-independent 16S rRNA amplicon sequencing (Illumina MiSeq platform) along with proximate composition analysis to characterise the bacterial community. The results showed that *Firmicutes* was the dominant phylum throughout fermentation, followed by *Proteobacteria*, indicating a microbial profile similar to that reported for other Asian fermented soybean foods.

Using culture-dependent isolation followed by whole-genome sequencing (Illumina Hi-Seq) and computational genome annotation, *Bacillus subtilis* Miz-8 was recovered from *Bekang* and compared with the reference strain *B. subtilis* subsp. *natto* BEST195 (Kumar et al., 2023). Genomic analysis revealed genes responsible for the biosynthesis of

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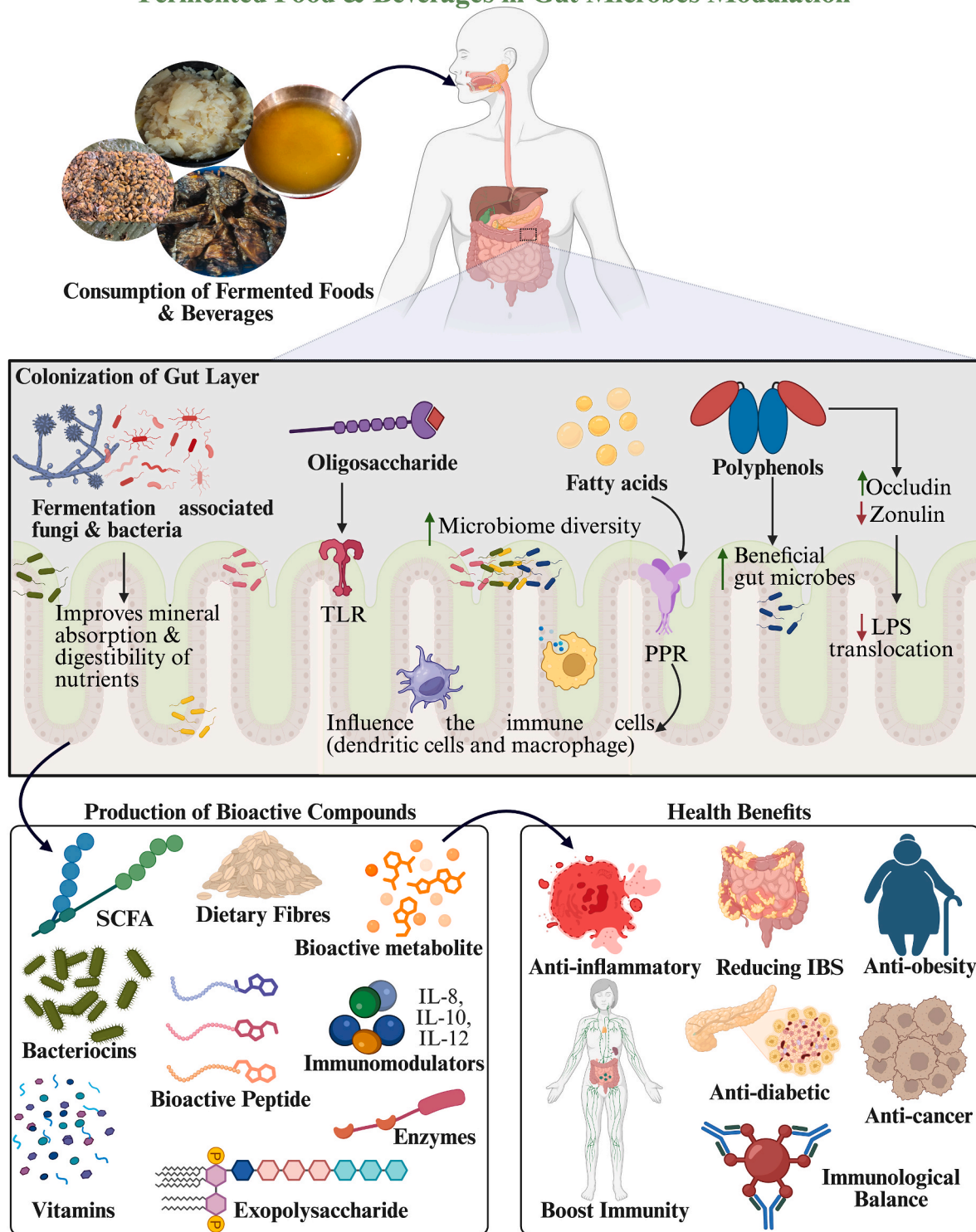


Fig. 5. Schematic representation illustrating the mechanistic pathways through which traditional fermented foods and beverages can modulate gut microbiota and confer health benefits. Consuming fermented foods introduces fermentation-associated beneficial bacteria and fungi, along with their bioactive metabolites, into the gastrointestinal tract. These microbes colonise the gut epithelium and enhance mineral absorption and nutrient digestibility. Dietary oligosaccharides, polyphenols, and fermentation-derived fatty acids support the proliferation of beneficial gut microbiota, positively modulate gut microbial composition, and activate pattern-recognition receptors (PPRs) and Toll-like receptors (TLRs), thereby influencing innate immune cells such as dendritic cells and macrophages. Fermentation-linked polyphenols also strengthen intestinal tight junctions (e.g., increased occludin and reduced zonulin expression) and limit lipopolysaccharide (LPS) translocation, contributing to improved gut barrier integrity. The microbial metabolic activity of fermented foods leads to the production of a wide range of bioactive molecules, including short-chain fatty acids (SCFAs), dietary fibre metabolites, vitamins, bioactive peptides, bacteriocins, exopolysaccharides, enzymes, and immunomodulatory metabolites. These compounds collectively regulate host immune responses, enhance mucosal immunity, and maintain gut microbial homeostasis.

Table 1

Comprehensive overview of traditional fermented foods and beverages from Northeast India and their impact on gut microbiota.

Source	Fermented Local Name	Food or Beverage	Isolated Microflora from Fermentation	Method used	Alcoholic or Non-alcoholic	North-eastern State of India	Key Findings	Reference
Soybean	Akhone/ Axone	Food	<i>B. coagulans</i> , <i>B. pentothenicus</i> , <i>B. subtilis</i> , <i>B. stearothermophilus</i> , <i>B. lentus</i> , <i>Ignatzschinaria</i> , <i>Corynebacterium</i> , <i>Oceanobacillus</i> , <i>E. caecorum</i> , <i>E. hirae</i> , <i>E. faecalis</i> , <i>S. xylosus</i> , <i>S. epidermidis</i> , <i>S. sciur</i> , <i>Aeromonas</i> , <i>Citrobacter</i> , <i>Enterobacter</i> , <i>Erwinia</i> , <i>Hafnia</i> , <i>Klebsiella</i> , <i>Morganella</i> , <i>Providencia</i> , <i>Proteus</i> , <i>Pseudomonas</i> , <i>Salmonella</i>	Microbial isolation, metabolite profiling	Non-alcoholic	Nagaland	Fermentation enhances the production of bioactive microbial metabolites that contribute to improved intestinal ecology and metabolic regulation.	Das et al. (2025)
		Food	<i>Bacillus</i> , <i>Enterococcus</i> , <i>Staphylococcus</i> , <i>E. coli</i> , <i>Aeromonas</i> , <i>Citrobacter</i> , <i>Enterobacter</i> , <i>Erwinia</i> , <i>Hafnia</i> , <i>Klebsiella</i> , <i>Morganella</i> , <i>Providencia</i> , <i>Proteus</i> , <i>Pseudomonas</i> , <i>Salmonella</i>	Culture, biochemical, microbial profiling	Non-alcoholic	Nagaland	Microbial community dominated by <i>Bacillus</i> species; presence of coliforms and <i>enterococci</i> suggests potential for gut microbial modulation.	Singh et al. (2014)
		Food	<i>Firmicutes</i> , <i>Proteobacteria</i> , <i>Staphylococcus</i>	16S rRNA amplicon sequencing	Non-alcoholic	Mizoram	Associations between <i>Staphylococcus</i> and macronutrient components suggest microbial interactions that may influence gut nutrient availability and microbial ecology.	Das et al. (2023) ; Deka et al. (2021)
	Bekang	Food	<i>B. subtilis</i>	Whole-genome sequencing	Non-alcoholic	Mizoram	Secondary metabolite production by <i>B. subtilis</i> Miz-8 supports beneficial microbial modulation without horizontal transfer of antimicrobial resistance genes, ensuring safety.	Kumar et al. (2023)
		Food	<i>B. subtilis</i> , <i>B. licheniformis</i> , <i>B. cereus</i> , <i>S. aureus</i> , <i>S. sciuri</i>	PCR, 16S rDNA, RFLP	Non-alcoholic	Manipur	High genetic variability in <i>B. subtilis</i> populations highlights potential for strain selection to develop standardized starter cultures with gut-modulatory effects.	Jeyaram et al. (2008)
	Hawaijar	Food	<i>B. subtilis</i> , <i>B. cereus</i> , other <i>Bacillus</i> species	Culture-dependent isolation	Non-alcoholic	Manipur	<i>Bacillus</i> -driven starch hydrolysis enhances substrate bioavailability, facilitating colonic microbial fermentation and gut microbial support.	Premarani Thingom & Chhetry, 2008
		Food	<i>B. subtilis</i> , <i>B. amyloliquefaciens</i> , <i>B. licheniformis</i> , <i>B. pumilus</i> , <i>Actinobacteria</i> , <i>C. glutamicum</i> , <i>L. lactis</i> , <i>P. mirabilis</i> , <i>S. aureus</i> , <i>P. penneri</i> , <i>E. faecalis</i> , <i>S. saprophyticus</i> , bacteriophages	-	Non-alcoholic	Sikkim	<i>Bacillus</i> and LAB-derived hydrolytic and decarboxylase enzymes generate health-promoting metabolites that enhance microbial diversity in the gut.	Sanjukta et al. (2022)
	Kinema	Food	<i>B. subtilis</i> , <i>B. amyloliquefaciens</i> , <i>B. licheniformis</i> , <i>B. pumilus</i> , <i>C. glutamicum</i> , <i>L. lactis</i> , <i>B. cereus</i> , <i>P. mirabilis</i> , <i>S. aureus</i> , <i>Proteus penneri</i> , <i>E. faecalis</i> , <i>B. tequilensis</i>	Metagenomic sequencing	Non-alcoholic	Sikkim	Seasonal microbial dynamics influence enzyme-mediated bioactive production, leading to modulation of beneficial gut taxa.	Kumar et al. (2019)
		Food	<i>B. tequilensis</i>	Biosurfactant extraction, lipopeptide analysis	Non-alcoholic	Sikkim	Lipopeptide biosurfactants synthesized by <i>B. tequilensis</i> exhibit potential in supporting the growth of beneficial gut microorganisms.	Chaurasia et al. (2020)

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Table 1 (continued)

Source	Fermented Local Name	Food or Beverage	Isolated Microflora from Fermentation	Method used	Alcoholic or Non-alcoholic	North-eastern State of India	Key Findings	Reference
	<i>Tungrymbai</i>	Food	<i>B. subtilis</i> , <i>B. licheniformis</i> , <i>B. pumilus</i> , <i>Enterococcus</i> , <i>L. fermentum</i> , <i>L. plantarum</i>	-	Non-alcoholic	Meghalaya	LAB display probiotic functionality including bile tolerance, cholesterol assimilation, and pathogen suppression, thereby enriching gut microbial diversity.	Hati et al. (2024)
		Food	<i>B. subtilis</i> , <i>B. licheniformis</i> , <i>B. pumilus</i>	ARDRA, ITS-PCR and RAPD-PCR techniques	Non-alcoholic	Meghalaya	<i>Enterococcus</i> species demonstrate probiotic activity, contributing to gut microbial modulation through antagonism of pathogens and metabolic regulation.	Chettri & Tamang, 2015
		Food	<i>L. fermentum</i> , <i>L. plantarum</i>	16S rRNA, 23S rRNA	Non-alcoholic	Meghalaya	Controlled LAB fermentation enhances nutrient bioavailability and sensory properties, augmenting gut-modulatory potential.	Mishra et al. (2017)
		Food	<i>L. fermentum</i> , <i>L. plantarum</i>	16S rRNA	Non-alcoholic	Meghalaya	Post-cooking addition of fermented soybean improves organoleptic quality and introduces bioactive compounds with potential gut health benefits.	Malakar et al. (2017)
		Food	Lactic acid bacteria (LAB)	Culture-dependent isolation	Non-alcoholic	Meghalaya	High LAB density and continued acid production suppress pathogens while sustaining beneficial gut taxa, thereby supporting microbial homeostasis.	Basaawmoit & Mishra, 2018
Bamboo	<i>Khorisa</i>	Food	<i>L. plantarum</i> , <i>L. brevis</i> , <i>L. paracasei</i> , <i>L. pentosus</i> , <i>L. collinoides</i>	16S rRNA sequencing	Non-alcoholic	Assam	LAB produce organic acids and bacteriocins, lowering luminal pH, suppressing pathogens, and promoting beneficial gut taxa.	Phovisay et al. (2024)
		Food	<i>Lactobacillus</i>	Antimicrobial assay	Non-alcoholic	Assam	<i>Lactobacillus</i> produces antibacterial secondary metabolites, particularly against <i>Staphylococcus aureus</i>	Kumar et al. (2022)
		Food	<i>Bacillus</i> , <i>Paenibacillus</i> , <i>Oceanobacillus</i> , <i>Lactobacillus</i> , <i>Enterococcus</i>	Culture-dependent isolation	Non-alcoholic	Assam	<i>Bacillus</i> produces extracellular enzymes (amylases, proteases, lipases), enhancing nutrient bioavailability and supporting gut microbial growth.	Asifa Qureshi et al., 2016
		Food	<i>L. plantarum</i> , <i>L. brevis</i> , <i>L. paracasei</i> , <i>L. pentosus</i> , <i>L. collinoides</i>	16S rRNA sequencing, GC-MS	Non-alcoholic	Assam	LAB produce bacteriocins targeting pathogens while sustaining beneficial gut commensals.	Sharma & Barooah, 2017
		Food	<i>Lactobacillus</i>	Culture-dependent isolation	Non-alcoholic	Assam	LAB exhibit probiotic traits including acid/bile tolerance and adhesion, promoting gut colonisation and pathogen inhibition.	Borthakur et al. (2022)
	<i>Soidon/ Soibum</i>	Food	<i>W. cibaria</i> , <i>W. ghanensis</i> , <i>L. lactis</i> , <i>L. citreum</i> , <i>L. brevis</i> , <i>L. plantarum</i>	PCR-DGGE, sequencing	Non-alcoholic	Manipur	LAB-driven ecological succession promotes acidification and bacteriocin production, enhancing colonisation resistance and gut microbial diversity.	Romi et al. (2015)
		Food	<i>B. subtilis</i> , <i>B. cereus</i> , <i>L. plantarum</i> , <i>Carnobacterium</i>	Culture enzyme assay	Non-alcoholic	Manipur	LAB and <i>Bacillus</i> produce antimicrobial metabolites, providing competitive advantages	Jeyaram et al. (2010)

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Table 1 (continued)

Source	Fermented Local Name	Food or Beverage	Isolated Microflora from Fermentation	Method used	Alcoholic or Non-alcoholic	North-eastern State of India	Key Findings	Reference
	Rhujuk/ Bastanga	Food	<i>B. subtilis</i> , <i>B. cereus</i> , <i>L. brevis</i>	PCR, 16S rDNA sequencing,	Non-alcoholic	Manipur	against pathogens and supporting gut microbial modulation. Genetic diversity in <i>Bacillus</i> and <i>Lactobacillus</i> strains supports antioxidant activity and reduced cyanide content, promoting gut health.	Hoikhokim & Kananbala, 2016
		Food	<i>B. subtilis</i> , <i>B. cereus</i> , <i>L. brevis</i>	DNA marker analysis	Non-alcoholic	Manipur	DNA markers linked to antioxidant activity and reduced cyanide levels support gut microbial homeostasis.	Khunjan et al. (2017)
		Food	<i>L. plantarum</i> , <i>E. faecium</i>	PCR, 16S rDNA sequencing	Non-alcoholic	Manipur	LAB exhibit probiotic traits, promotes beneficial taxa and gut stability.	Khunjan and Pandey (2018)
		Food	<i>B. safensis</i> , <i>B. tequilensis</i> , <i>B. siamensis</i> , <i>B. nakamurai</i> , <i>B. subtilis</i>	Metagenomic sequencing	Non-alcoholic	Nagaland	<i>Bacillus</i> strains exhibit antibacterial activity against pathogens, promotes gut health and microbial modulation.	Das, Afzal, et al. (2023)
		Food	<i>B. safensis</i> , <i>B. tequilensis</i> , <i>B. siamensis</i> , <i>B. nakamurai</i> , <i>B. subtilis</i>	Metagenomic sequencing	Non-alcoholic	Nagaland	High phenolic content and antioxidant activity support beneficial GM	Jamir and Deb (2021)
		Food	<i>L. plantarum</i> , <i>L. brevis</i> , <i>P. pentosaceus</i>	Culture-dependent isolation	Non-alcoholic	Sikkim	LAB succession drives acidification, enhancing gut microbial modulation through organic acid production and pathogen suppression.	Tamang and Sarkar (1996)
Rice	Poita Bhat	Food	<i>W. confusa</i>	Culture +16S rDNA	Non-alcoholic	Assam	Exhibits probiotic traits, supporting gut colonisation and pathogen suppression.	Nath et al. (2021)
Fish	Hentak	Food	<i>L. pentosus</i>	Culture, antimicrobial assay	Non-alcoholic	Manipur	<i>L. pentosus</i> produces antibacterial substances, supporting gut colonisation and pathogen suppression.	Aarti et al. (2016)
		Food	<i>L. brevis</i>	Culture-dependent isolation followed by phenotypic and biochemical characterization of LAB	Non-alcoholic	Manipur	<i>L. brevis</i> exhibits probiotic and antioxidant properties, supporting gut microbial stability.	Aarti et al. (2017)
	Namsing	Food	<i>Bacillus</i> , <i>Lactobacillus</i> , <i>Kurthia</i> , <i>Providencia</i> , <i>Wohlfahrtiimonas</i> , <i>Paenibacillus</i> , <i>Staphylococcus</i> , <i>Klebsiella</i> , <i>Vagococcus</i>	Culture-dependent microbial isolation combined with biochemical identification of bacterial communities	Non-alcoholic	Assam	Fermentation enhances mineral bioavailability and digestibility, supporting gut health via probiotic activity.	Baruah et al. (2021)
		Food	<i>Bacillus</i> , <i>Lactobacillus</i> , <i>Kurthia</i> , <i>Providencia</i> , <i>Wohlfahrtiimonas</i> , <i>Paenibacillus</i> , <i>Staphylococcus</i> , <i>Klebsiella</i> , <i>Vagococcus</i>	Culture-dependent isolation followed by probiotic property evaluation and biochemical identification	Non-alcoholic	Assam	LAB and <i>Bacillus</i> contribute to gut health through protease activity and pathogen inhibition.	Chakrabarty et al., 2019
Pig Fat	Sathu	Food	<i>Bacillus</i> , <i>Lactobacillus</i>	Culture-dependent isolation	Non-alcoholic	Assam	<i>Bacillus</i> and <i>Lactobacillus</i> exhibits probiotic traits (acid/bile resistance), supporting gut microbial stability.	Yukhaibam and Lhouvum (2021)
		Food	<i>Bacillus</i> , <i>Lactobacillus</i>	Culture-based methods followed by morphological and biochemical characterization	Non-alcoholic	Assam	Probiotic properties of <i>Bacillus</i> and <i>Lactobacillus</i> enhance gut health through microbial modulation.	Chakrabarty et al. (2014)
Plants	Kharoli/ Kahudi	Food	<i>S. succinus</i> , <i>S. saprophyticus</i> , <i>Bacillus subtilis</i> , <i>Weissella</i> , <i>Corynebacterium</i> , <i>Aerococcus</i> ,	Multi-omics analysis	Non-alcoholic	Assam	Core species enhance PUFA content, modulate genes related to gut homeostasis, barrier	Das et al. (2024); Ojha et al. (2023)

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Table 1 (continued)

Source	Fermented Local Name	Food or Beverage	Isolated Microflora from Fermentation	Method used	Alcoholic or Non-alcoholic	North-eastern State of India	Key Findings	Reference
	Gundruk	Food	<i>Ruoffia</i> , <i>Paenibacillus</i> , <i>Enterococcus</i> <i>Lactobacillus spicheri</i> G2	Culture-dependent isolation followed by 16S rRNA gene sequencing	Non-alcoholic	Sikkim	integrity, and immune regulation. A novel bacteriocin-producing strain (<i>L. spicheri</i> G2) modulated GM by inhibiting pathogens, promoting beneficial bacteria, enhancing SCFA production, and supporting gut barrier integrity and metabolic regulation.	Gautam and Sharma (2015)
Alocoholic Beverage	Atingba	Beverage	<i>L. plantarum</i>	Antibiofilm assay, and pathogen inhibition assay	Alcoholic	Manipur	<i>L. plantarum</i> produces bacteriocins with antimicrobial activity, potentially enriching beneficial gut bacteria.	Loying et al. (2024)
			<i>L. plantarum</i>	Biofilm inhibition assay	Alcoholic	Manipur	<i>L. plantarum</i> produces bacteriocins, inhibiting pathogens and biofilms, supporting gut homeostasis.	Huidrom et al. (2024)
	Chubitchi	Beverage	<i>L. brevis</i>	16S rDNA + biochemical	Alcoholic	Meghalaya	<i>L. brevis</i> exhibits probiotic traits (acid/bile tolerance, hydrophobicity, auto-aggregation), facilitating gut colonisation and pathogen inhibition.	Murugan & Mishra, 2022
	Apong	Beverage	<i>L. plantarum</i> , <i>L. lactis</i> , <i>L. brevis</i> , <i>L. mesenteroides</i> , <i>W. paramesenteroides</i> , <i>L. kimchii</i> , <i>P. pentosaceus</i> , <i>L. casei</i>	Metagenomic sequencing	Alcoholic	Assam	Increases gut microbial diversity, enriches specific bacterial genera (e.g., <i>Eubacterium ventriosum</i> , <i>Howardella</i>)	Deka et al. (2025)
	Chokot	Beverage	<i>Bacillus</i> , <i>Arthrobacter</i> , <i>Lactobacillus</i> , <i>Ilyobacter</i> , <i>Clostridium</i> , <i>Lactococcus</i>	Metagenomic analysis, GC-MS metabolite profiling	Alcoholic	Assam	Diverse microbial population produces antimicrobial and antifungal metabolites, potentially modulating GM.	Bhattacharjee et al. (2023)
	Xaj-pani	Beverage	<i>S. cerevisiae</i>	Microbial isolation, culture identification	Alcoholic	Assam	Enriches beneficial gut bacteria and supports SCFA production, enhances intestinal barrier function.	Goswami et al. (2020)
			<i>S. cerevisiae</i>	Microbial characterization and fermentation analysis	Alcoholic	Assam	Promotes microbial diversity and SCFA production	Fuloria et al. (2022)

several secondary metabolites, including surfactin, fengycin, bacillibactin, bacilysin, subtilisin A, and α -amylase, which are known to enhance fermentation, inhibit spoilage microorganisms, and exhibit health-promoting activities. The predominance of *Bacillus* and *Lactobacillus spp.* suggests their involvement in protein degradation, antimicrobial metabolite production, and possible probiotic effects, supporting the functional importance of *Bekang* fermentation.

Safety evaluation indicated that although Miz-8 possesses virulence genes similar to those of BEST195, it is considered safe for human consumption (Kumar et al., 2023). The strain contains a single intact prophage region, but the absence of the integrase protein prevents horizontal transfer of antimicrobial resistance genes. Among the predicted resistance genes, only *tmrB* was detected, and the plasmid lacked both prophage sequences and resistance determinants, indicating a very low risk of antibiotic resistance dissemination. In addition, the presence of several single-nucleotide polymorphisms and 344 insertion-deletion (indel) variations demonstrates the genetic uniqueness of the strain, further supporting its safe use in traditional fermentation.

3.1.3. Hawaijar

Hawaijar is a traditional fermented soybean delicacy from Manipur, valued for its cultural relevance and distinctive culinary attributes. It exhibits an alkaline, sticky, and mucilaginous texture, along with a mildly pungent flavour, comparable to Southeast Asian fermented soybean products such as *Natto* and *Thua nao* (Singh et al., 2023). To characterise its indigenous microbial community, Jeyaram et al. (2008) analysed *Hawaijar* prepared in traditional households in Manipur using PCR amplification of the 16S-23S rDNA spacer region, followed by RFLP analysis and sequencing of the 9-1514 region of the 16S rDNA. Their findings revealed three predominant phylogenetic groups: the *Bacillus subtilis* group (including *B. subtilis* and *B. licheniformis*), comprising 112 isolates; the *Bacillus cereus* group, comprising 128 isolates; and the *Staphylococcus* group (including *S. aureus* and *S. sciuri*), comprising 23 isolates. Additionally, a small number of isolates were identified as *Alcaligenes spp.* and *Providencia rettgeri* (Jeyaram et al., 2008).

In another study, 26 nos. of *Hawaijar* samples were collected from five districts of the Imphal Valley, yielding 152 *Bacillus* strains (Premarani Thingom & Chhetry, 2008). The isolates were first sorted by



Fig. 6. Images of selected traditional fermented foods prepared across different states of Northeast India. Fermented Soybean Based: A: *Akhuni* (Nagaland), B: *Bekang* (Mizoram), C: *Tungrymbai* (Meghalaya), Fermented Bamboo Based: D: *Khorisa* (Assam), Fermented beverages: E: *Temi Opo* (Arunachal Pradesh), F: *Sai Mod* (Assam), Fermented Rice Based: G: *Poita Bhat* (Assam), Fermented Meat/Fish Based: H: *Sukuti* (Assam), I: *Shidal* (Tripura), J: *Ngari* (Manipur), K: *Sa-um* (Mizoram), Fermented Meat/Fish Based: L: *Kharoli* (Assam), M: *Gundruk* (Sikkim), N: *Anishi* (Nagaland), O: *Buratamul* (Assam), Fermented Milk Based: P: *Doi* (Assam).

colony morphology and cell characteristics, and representative strains from each group were further analysed by PCR-RAPD profiling. Molecular identification using the 16S-23S rDNA spacer region followed by restriction digestion identified the isolates as *B. subtilis*, *B. cereus*, as well as other *spp.* of *Bacillus*, as per MTCC type strains. Functional characterisation identified amylase activity throughout the *Bacillus* isolates, whereas protease and lipase activities were relatively weaker, indicating their significant involvement in starch degradation during fermentation (Premarani Thingom & Chhetry, 2008). *Hawaijar* hosts a complex microbial ecosystem dominated by beneficial *Bacillus spp.*, particularly *B. subtilis* and *B. licheniformis*, which play crucial roles in starch degradation, the production of bioactive compounds, and flavour development, thereby contributing to the functional properties of fermented foods, as illustrated in Fig. 5. These beneficial strains are being evaluated as potential starter cultures to achieve controlled and standardised *Hawaijar* fermentation while preserving its nutritional, functional, and sensory integrity (Jeyaram et al., 2008).

However, the presence of *B. cereus* and *S. aureus* highlights potential safety concerns, as these *spp.* can act as opportunistic pathogens under certain conditions. Their detection indicates that although traditional fermentation promotes beneficial microbial activity, inadequate hygiene or uncontrolled fermentation conditions may allow undesirable microorganisms to survive, emphasising the need for improved standardisation and safety monitoring during preparation.

3.1.4. Kinema

Kinema is a fermented soybean food commonly eaten in the eastern Himalayan states, particularly in Sikkim. It is prepared through the natural fermentation of boiled soybeans, primarily facilitated by *B. subtilis* and associated *spp.*, which give it a typical sticky consistency, pungent smell, and distinctive flavour. Its preparation and consumption are essential to the cultural and food heritage of the people of the Himalayas (Sanjukta et al., 2022).

Kumar et al. (2019) performed a metagenomic analysis of *Kinema* samples collected from a local market in Gangtok, Sikkim, to investigate the microbial diversity and functional potential of this fermented soybean product across different seasons. The results showed that *Firmicutes* was the dominant phylum, followed by *Actinobacteria* and *Proteobacteria*. Among the major *spp.* identified, *Bacillus spp.* were the most prevalent, along with *Corynebacterium glutamicum* and *Lactococcus lactis*. Seasonal variation significantly influenced the microbial community structure. The study also reported the occasional presence of undesirable microorganisms, including *Bacillus cereus*, *Proteus mirabilis*, *S. aureus*, *Proteus penneri*, *Enterococcus faecalis*, and *Staphylococcus saprophyticus*, during certain seasons, highlighting the dynamic nature of *Kinema* fermentation. In addition, bacteriophages belonging to the families *Siphoviridae*, *Myoviridae*, and *Podoviridae* were detected, indicating a complex microbial ecosystem. Functional annotation revealed a diverse set of enzymes involved in protein and carbohydrate degradation, including β -galactosidase, β -xylosidase, β -glucosidase, and glutamate

decarboxylase, which contribute to the formation of beneficial metabolites. Overall, *Kinema* fermentation is primarily driven by beneficial *Bacillus* spp., which promote starch and protein hydrolysis, produce bioactive compounds, and contribute to the characteristic texture, aroma, and flavour of the product.

3.1.5. *Tungrymbai*

Tungrymbai is an ethnic fermented soybean product widely consumed by the Khasi and Garo tribes of Meghalaya and is a staple of the local diet. The product is produced via spontaneous fermentation of soybeans, primarily mediated by diverse populations of LAB and *Bacillus* spp. (Hati et al., 2024). The typical appearance and traditional form of this fermented food is illustrated in Fig. 6, highlighting its coarse texture and darkened colour resulting from fermentation.

Microbiological characterisation of various *Tungrymbai* samples collected from villages and markets in Meghalaya revealed high microbial loads, dominated primarily by *B. subtilis*, *B. licheniformis*, and *B. pumilus*, which were identified as the major contributors to fermentation (Chettri & Tamang, 2015). Additionally, LAB, especially *Enterococcus* spp. isolated from *Tungrymbai*, were detected as part of the indigenous fermentation microbiota. These LAB isolates showed promising probiotic attributes, including the ability to withstand simulated gastric acidity and bile salts, susceptibility to commonly used antibiotics, and cholesterol-lowering activity. They also effectively inhibited the growth of pathogenic bacteria, including *S. aureus* and *E. coli*, indicating their potential to promote gut health and improve food safety (Chettri & Tamang, 2015). Beyond traditional household fermentation, controlled laboratory fermentation with well-characterised LAB strains, such as *Lactobacillus fermentum* and *Lactobacillus plantarum*, obtained through culture-dependent isolation and strain identification using conventional microbiological and molecular methods, has been shown to improve the quality of fermented soy products (Mishra et al., 2017). Inoculation with 1% biomass, followed by incubation at 37 °C for 3–4 days, resulted in enhanced sensory characteristics, including improved aroma, taste, texture, colour, and overall acceptability, compared with naturally fermented products. Incorporating fermented soy into cooked dishes further improved the sensory appeal, highlighting the influence of processing and culinary practices on the final product quality (Malakar et al., 2017).

Storage studies on *Tungrymbai* revealed that its functional properties remain dynamic over time, as evaluated using culture-dependent microbial enumeration (LAB plate count) along with physicochemical analysis, showing that during an eight-day storage period, LAB counts remained high, while the pH gradually declined to 5.34 and titratable acidity increased to 0.054% lactic acid (Basaiaiwmoit & Mishra, 2018). These changes indicate continuous microbial activity and biochemical transformation, contributing to the evolving quality and safety of the product during storage (Basaiaiwmoit & Mishra, 2018).

3.2. Fermented bamboo-based foods

3.2.1. *Khorisa*

Fermented bamboo shoot products derived from *Bambusa balcooa* in Northeast India serve not only as traditional dietary components but also as rich sources of functionally important microorganisms that may contribute to gut health (Sharma & Barooah, 2017; Phovisay et al., 2024). A wild-type *Lactobacillus* strain isolated from *Khorisa* collected from local villages in Manipur was shown to produce antibacterial secondary metabolites during submerged fermentation, particularly against *Staphylococcus aureus* (Kumar et al., 2022). Broader microbial profiling of fermented bamboo shoot products from Assam using 16S rRNA sequencing identified 44 bacterial variants, with *Bacillus*, *Oceanobacillus*, *Paenibacillus*, and *Lactobacillus* as the dominant genera (Asifa Qureshi, Itankar, Saikia, Mandal, & Purohit, 2016; Sharma & Barooah, 2017).

Detailed investigation of the LAB community in *Khorisa* revealed five predominant spp., *L. plantarum*, *L. pentosus*, *L. paracasei* subsp. *paracasei*,

L. brevis, and *L. collinoides*, with viable counts reaching up to 10⁷ CFU/mL (Sharma & Barooah, 2017). Borthakur et al. (2022) further analysed LAB isolated from *Khorisa* samples collected from different regions of Assam using culture-dependent isolation, biochemical and morphological characterisation, 16S rRNA sequencing, and GC-MS-based metabolite profiling, confirming the predominance of *Bacillus*, *Oceanobacillus*, *Paenibacillus*, and *Lactobacillus*, indicating a dynamic and cooperative fermentation ecosystem.

Functional assays *in-vitro* enzymatic activity screening showed that *Bacillus* isolates produce extracellular enzymes such as amylases, proteases, and lipases, which hydrolyse complex polysaccharides, proteins, and lipids into bioavailable forms, thereby enhancing nutritional quality and generating substrates such as oligosaccharides, peptides, and SCFA precursors that may support beneficial gut microbes (Sharma & Barooah, 2017). GC-MS analysis further confirmed the presence of organic acids, amino acids, esters, and vitamins, highlighting the multifactorial contribution of fermentation to gut health. LAB isolates exhibited antimicrobial activity against foodborne pathogens by producing organic acids and bacteriocins, which remained active across a wide range of pH, temperature, and enzymatic conditions, suggesting stability during gastrointestinal transit (Sharma & Barooah, 2017). These ribosomally synthesized peptides act by binding to specific receptors on the bacterial cell membrane, causing pore formation, membrane disruption, or interference with cell wall biosynthesis, leading to cell death, while beneficial commensal bacteria lacking these receptors remain largely unaffected (Yang et al., 2014). The ability of LAB to tolerate low pH, bile salts, and antimicrobial environments further supports their potential survival in the gastrointestinal tract and their role in inhibiting pathogenic bacteria (Simons et al., 2020).

Further characterization demonstrated that LAB strains isolated from *Khorisa* possess tolerance to acidic and alkaline conditions, indicating their suitability as probiotic candidates (Borthakur et al., 2022). *In vitro* adhesion assays confirmed their ability to attach to surfaces, an important property for intestinal colonisation. The combined activity of spore-forming *Bacillus* and *Paenibacillus* during early fermentation, followed by *Lactobacillus* dominance in later stages, contributes to substrate breakdown, bioactive metabolite production, and improved food quality (Borthakur et al., 2022).

Safety evaluation showed that the isolated LAB strains lacked hemolytic, catalase, and oxidase activity and displayed favorable antibiotic susceptibility profiles, supporting their potential probiotic safety (Borthakur et al., 2022). Previous reports on LAB from fermented bamboo products also indicated the absence of biogenic amine production (Tamang & Tamang, 2009). However, a comprehensive safety assessment should additionally include evaluation of transferable antibiotic resistance genes and other virulence-associated traits before confirming probiotic suitability. From a biochemical perspective, GC-MS-based metabolite profiling reveals a diverse array of compounds in fermented bamboo shoot products (Sharma & Barooah, 2017; Borthakur et al., 2022). While such compositional complexity is suggestive of potential functional relevance, the mere presence of these metabolites does not establish their bioavailability or direct role in modulating GM upon consumption. Therefore, these findings should be interpreted cautiously and considered indicative of the potential of fermented bamboo shoot products as candidates for future mechanistic, *in vivo*, and clinical studies to validate their effects on the host GM and overall health.

3.2.2. *Soidon/soibum*

Bamboo shoot fermentation from the succulent shoot tip of *Bambusa tulda* represents a deeply entrenched culinary tradition in Northeast India, particularly in Manipur, where it forms an integral component of the local diet (Behera & Balaji, 2021; Andrew, 2022). The microbial ecology of bamboo shoot fermentation in *Soidon*, carried out by indigenous communities of Manipur, has been well documented using both cultivation-based and advanced molecular approaches (Romi et al.,

2015).

Techniques including PCR-amplified ribosomal DNA restriction analysis, rRNA gene sequencing, qPCR, PCR-DGGE, and Illumina amplicon sequencing have demonstrated a clear three-stage microbial succession during natural fermentation of bamboo shoots. During the initial 1-2 days, *Weissella* spp., particularly *W. cibaria* and uncultured *W. ghanensis*, together with *Lactococcus lactis* subsp. *Cremoris* dominates the microbial community. By the third day, *Leuconostoc citreum* becomes more prevalent, marking the intermediate stage, whereas in the later phase (5-7 days), *L. plantarum* and *Lactobacillus brevis* predominate as *Weissella* and *Lactococcus* decline (Romi et al., 2015). Similarly, *Soidon* prepared from tender shoots of *Schizostachyum capitatum* using the acidic starter culture “*Soidon mahi*” (pH ~4.5) contains a mixed microbial community dominated by *Bacillus* spp. and LAB, identified through culture-dependent and molecular methods, including ARDRA and 16S rDNA sequencing (Jeyaram et al., 2010). Frequently detected spp. include *B. subtilis*, *B. cereus*, *L. plantarum*, *Limosilactobacillus brevis*, and *Carnobacterium* spp., confirmed by culture-dependent isolation, ARDRA, 16S rRNA sequencing, and amplicon-based microbial profiling (Hoikhokim & Kananbala, 2016). Genetic analyses using ITS-RFLP and RAPD profiling revealed distinct clades within *B. subtilis*, *B. cereus*, and *L. brevis*, while AFLP-based genotyping of *Dendrocalamus hamiltonii* identified markers associated with enhanced antioxidant activity and reduced cyanide content (Khunjan et al., 2017).

The progressive microbial succession reflects gradual acidification, establishment of a stable fermentation environment, and enrichment of microorganisms with potential gut-health benefits (Romi et al., 2015). Fermentation also leads to significant biochemical transformations, producing metabolites of nutritional and pharmacological importance. Fermented shoots of *Bambusa vulgaris* and *Dendrocalamus strictus* used in *Soidon* contain phytosterols such as stigmasterol, which can be bio-transformed into androsta-1,4-diene-3,17-dione by *Arthrobacter globiformis* (Sarangthem et al., 1998). LAB-derived metabolites, including organic acids and bacteriocins, show inhibitory activity against pathogenic microorganisms both during fermentation and after consumption, potentially improving gastrointestinal health (Hoikhokim & Kananbala, 2016). Fermentation further enhances the bioavailability of essential vitamins such as vitamin B12, folate (B9), and vitamin C, and generates diverse volatile compounds that promote cross-feeding interactions within the gut microbiome, thereby increasing microbial diversity (Sonar et al., 2015). The probiotic potential of LAB isolated from fermented bamboo shoots has been confirmed through culture-dependent isolation, biochemical characterisation, and 16S rRNA identification (Khunjan & Pandey, 2018). Through the combined activity of LAB, *Bacillus* spp., and fermentation-derived metabolites, these foods may enhance microbial diversity, suppress pathogens, improve nutrient availability, and provide metabolic and immunological benefits (Gopal, 2023). Although the presence of *B. cereus* in some preparations raises safety concerns, the sustained dominance of LAB, particularly *L. plantarum* and *L. brevis*, contributes to colonisation resistance and the maintenance of a beneficial microbial ecosystem (Hoikhokim & Kananbala, 2016). Fermentation also plays a detoxifying role by reducing cyanogenic glycosides such as taxiphyllin, thereby lowering hydrogen cyanide levels to safe limits. This detoxification process further supports the safety, nutritional quality, and functional value of fermented bamboo shoot products as traditional foods (Khunjan et al., 2017).

3.2.3. Rhujuk/bastanga

Nagaland, known for its rich tribal culinary heritage, produces several vegetable-based TFF, including *Rhujuk* or *Bastanga*, prepared using *B. vulgaris* and *D. strictus* (Das et al., 2023). Fermentation is typically carried out at the household level through spontaneous microbial processes, leading to variations in sensory and nutritional qualities. Microbiological assessments have consistently reported the predominance of *Bacillus* spp. in these products, while occasional detection of

Staphylococcus highlights the importance of implementing better hygiene and controlled fermentation practices. Biochemical analyses indicate that *Rhujuk/Bastanga* from various Naga communities have a notable nutritional profile, being rich in protein, carbohydrates, fibre, and essential minerals, while remaining low in fat (Dey et al., 2023). This product also exhibits elevated levels of total phenolics, which align with its potent antioxidant activity and suggest potential health benefits such as mitigating oxidative stress and supporting the growth of beneficial gut microbes (Jamir & Deb, 2021). 16S rRNA gene sequencing revealed a diverse bacterial composition dominated by *Bacillus* taxa, including *B. safensis*, *B. tequilensis*, *B. siamensis*, *B. nakamurai*, and *B. subtilis*, along with *Enterobacter* spp., many of which had not been previously associated with fermented bamboo shoot products (Jamir & Deb, 2021). Also, the absence of *Lactobacillus* strains or moulds, which commonly drive fermentation in other traditional foods, highlights the distinct and unique microbial ecology of these products. Moreover, the discovery of novel *Bacillus* strains exhibiting broad-spectrum antibacterial activity points towards the potential for natural food preservation.

3.2.4. Mesu

In Sikkim, *Mesu*, prepared from the young shoots of *D. hamiltonii*, has long been traditionally used both as a pickle and as a base for curries. Microbiological studies of 30 *Mesu* samples collected from local markets in Sikkim isolated 327 LAB strains, predominantly *L. plantarum*, *L. brevis*, and *Pediococcus pentosaceus* (Tamang & Sarkar, 1996). All three spp. were consistently detected in raw bamboo shoots, with *L. plantarum* emerging as the dominant spp., followed by *L. brevis*, while *P. pentosaceus* was recovered less frequently, from only 40-50% of samples. Fermentation followed a defined microbial succession, initiated by *P. pentosaceus*, followed by *L. brevis*, and ultimately dominated by *L. plantarum*. This succession was accompanied by significant biochemical changes, including an increase in titratable acidity from 0.04% to 0.95% and a concurrent decline in pH from 6.4 to 3.8, emphasising the role of LAB in acidification, preservation, and potential modulation of gut microbial communities (Tamang & Sarkar, 1996).

3.2.5. Other fermented bamboo food products

In Mizoram, these foods are produced using culturally inherited, largely uncontrolled fermentation methods, in which microbial diversity influences flavour, texture, and nutritional quality (Deka, Mehetre, et al., 2021). High-throughput sequencing of samples collected on the 3rd, 5th, and 7th days of fermentation showed that *Firmicutes* and *Proteobacteria* were the dominant phyla. At the genus level, *Lactobacillus*, *Staphylococcus*, and *Clostridium* were most abundant in *Bekang* and *Sa-um*, with their relative distribution changing during fermentation. *Lactobacillus* showed a positive association with fermented bamboo shoots, whereas *Staphylococcus* correlated with protein, carbohydrate, and fibre content in soybean products, indicating that substrate type and preparation method significantly influence microbial composition (Deka, Handique, et al., 2021; Deka, Mehetre, et al., 2021).

In Tripura, naturally fermented bamboo shoot products such as *Moiyapansung*, *Mileyeamileye*, *Moiyakoshak*, and *Midukeye* were analysed using shotgun metagenomic sequencing, revealing a highly diverse microbiome comprising 49 phyla, 409 families, 841 genera, and 1799 spp. (Das et al., 2023). *Firmicutes* dominated the community, followed by *Proteobacteria*, *Bacteroidetes*, *Actinobacteria*, and *Fusobacteria*. At the spp. level, *L. plantarum* was most prevalent, along with *Lactococcus lactis*, *Limosilactobacillus brevis*, *Leuconostoc mesenteroides*, *Weissella paramesenteroides*, *Leuconostoc kimchii*, *Pediococcus pentosaceus*, *Leuconostoc gasicomitatum*, and *Lactocaseibacillus casei*. Functional metagenomic analysis identified genes involved in amino acid and lipid metabolism, pectin degradation, and other pathways that contribute to improved nutritional value, flavour development, and sensory quality (Das, Tamang, et al., 2023).

Bamboo shoots such as *Bambusa bambos* (L.) Voss are rich in nutrients and bioactive compounds, including phytosterols, as demonstrated

by lipid extraction, sterol fraction isolation, and GC-MS-based profiling, which showed that boiling, soaking, and fermentation increase sterol concentration, with fermentation producing the greatest effect (Saini et al., 2024). Microbial community analysis using both culture-dependent and molecular approaches, including 16S rRNA sequencing and amplicon profiling, revealed the dominance of Firmicutes, especially *Lactobacillus* and related LAB, while Proteobacteria and other minor phyla varied with substrate and processing conditions (Das, Tamang, et al., 2024). Fermentation also increases the concentration and bioavailability of bioactive compounds such as phytosterols. Overall, these fermented foods represent complex microbial ecosystems that shape their nutritional, sensory, and functional characteristics, supporting their potential role in modulating the GM.

3.3. Fermented traditional beverages

Traditional fermented beverages of Northeast India are prepared using rice or other cereal substrates through natural fermentation involving mixed microbial consortia. Representative appearances of some commonly consumed fermented beverages are shown in Fig. 6E & F, illustrating the diversity in their colours, consistencies, and traditional preparation methods.

3.3.1. Apong

Apong is a traditional rice-based fermented alcoholic beverage of the *Mishing* community in India, prepared using cooked rice and a dried starter culture known as *Apong kuri*. During fermentation, LAB metabolise carbohydrates and proteins in the rice matrix, producing ethanol, organic acids, and a range of bioactive metabolites that impart the characteristic flavour and mild alcohol content of the drink (Deka et al., 2025). This process also fosters the development of a diverse microbial consortium that can beneficially influence the gut microbiome (Deka, Handique, & Deka, 2021).

Studies indicate that *Apong* consumption enhances gut microbial diversity and selectively enriches certain bacterial genera, with community composition varying by *Apong* type. Drinkers of *Nogin Apong* (made from steamed rice) exhibited higher abundances of *Howardella*, the *Eubacterium ventriosum* group, and certain *Bacteroidales*. In contrast, those consuming *Poro Apong* (made from roasted rice) showed enrichment of *Agathobacter*, *Oscillibacter*, and *Turicibacter* (Das, Özkurt, et al., 2023). The prominence of carbohydrate and amino acid-utilising bacteria, such as *Bacteroides* and *Lachnospira*, reflects both the nutritional composition of *Apong* and the selective pressures exerted during fermentation. These microbial communities not only contribute to fermentation and flavour development but also support gut health by promoting microbial diversity and increasing the abundance of beneficial taxa. Also, *Apong* intake has been associated with reduced levels of carboxylic acids such as succinate, without adversely affecting key SCFA essential for intestinal health. Such modulation of GM and metabolic intermediates suggests that regular *Apong* consumption may help maintain a balanced gut ecosystem, improve nutrient metabolism, and support overall health by producing bioactive fermentation-derived metabolites (Das, Özkurt, et al., 2023).

3.3.2. Atingba

Atingba, also known as *Waiyu*, is a traditional rice-based alcoholic beverage of Manipur, prepared from glutinous rice and local herbs, widely consumed by the Meitei community and recognised for its strong, vodka-like aroma and flavour (Loying et al., 2024).

Samples collected from the Andro and Seijang regions of Manipur have demonstrated antioxidant activity, effectively inhibiting lipid peroxidation and the growth of pathogenic bacteria, including *S. aureus* and *E. coli* (Mangang et al., 2017). When incorporated into rice beer preparation and stored at 32 °C for up to three months, *Atingba* shows minimal changes in acidity, colour, anthocyanin levels, and peroxide value, while substantially reducing microbial spoilage and eliminating

biogenic amines, indicating enhanced product stability and safety. Microbial profiling reveals a diverse community dominated by LAB, particularly *L. plantarum* (Huidrom et al., 2024). *In vitro* studies further confirm its bactericidal effects against methicillin-resistant *S. aureus* and its capacity to disrupt biofilm formation, highlighting its potential to support gastrointestinal health through microbial modulation. The functional attributes of *Atingba* are largely attributed to its predominant strain, *L. plantarum* BRD3A, which harbours multiple bacteriocins and secondary metabolite gene clusters that confer pathogen inhibition and improve shelf life.

3.3.3. Chokot

Choko/Chokot is a traditional alcoholic beverage prepared from local rice by the *Rabha* tribe of Assam, India (Yumnam et al., 2024). This beverage accounts for a substantial portion of the daily caloric intake for the *Rabha* community.

The microbial composition and biochemical characteristics of *Chokot* collected from a *Rabha* village were investigated by Bhattacharjee et al. (2023) using metagenomic analysis of the starter culture. The study revealed a diverse microbial consortium dominated by bacterial genera such as *Bacillus*, *Arthrobacter*, *Lactobacillus*, *Ilyobacter*, *Clostridium*, and *Lactococcus*. The presence of *Lactobacillus* and *Lactococcus* suggests that LAB are involved in fermentation and acidification, whereas *Bacillus* spp. may contribute to the enzymatic degradation of complex substrates. The detection of anaerobic genera such as *Clostridium* and *Ilyobacter* indicates that fermentation proceeds under partially anaerobic conditions, which is typical for traditional rice-based alcoholic fermentations. This diverse microbial community reflects the complexity of the starter culture used in indigenous fermentation practices.

Metabolite profiling of *Chokot* using GC-MS revealed several bioactive compounds with antimicrobial and antifungal activities (Bhattacharjee et al., 2023). These metabolites may contribute to the beverage's preservation and to potential health-related effects when consumed in moderate amounts. The coexistence of LAB and other fermentative microbes suggests that *Chokot* may also contain metabolites that support gut microbial balance, although further functional studies are required to confirm these effects.

3.3.4. Chubitchi

Fermented rice beverages are among the most widely consumed ethnic alcoholic drinks in Meghalaya, known as *Chubitchi* among the Garo tribe, prepared from non-glutinous rice (Marak et al., 2021). Microbial exploration of these traditionally prepared beverages has revealed a rich diversity of fermenting microbes. For instance, six yeast strains were isolated from twenty indigenous fermented food samples collected from the Garo Hills region (Mishra et al., 2018).

Building upon these findings, Murugan and Mishra (2022) investigated the isolation and characterisation of LAB from *Chubitchi* to determine their suitability as functional starter cultures with probiotic potential. The LAB isolates were identified through morphological and biochemical characterisation, followed by molecular identification using 16S rDNA sequencing and computational classification. The probiotic attributes of the isolates were systematically evaluated using a series of *in vitro* assays, including tolerance to acidic pH and bile salts, auto-aggregation, cell-surface hydrophobicity, antibacterial activity and antibiotic susceptibility testing. Among the identified strains, *L. brevis* showed the most promising probiotic characteristics, exhibiting high tolerance over a broad pH range and at elevated bile salt concentrations. In addition, its hydrophobicity and auto-aggregation properties suggest an enhanced capacity for adhesion to intestinal epithelial cells, which may facilitate gut colonisation and contribute to modulation of the intestinal microbiota.

Most isolates exhibited broad-spectrum antibacterial activity, particularly against Gram-positive pathogens, while retaining susceptibility to commonly used antibiotics, including ampicillin, rifampicin, gentamicin, erythromycin, and tetracycline, which are important for

ensuring safety in probiotic applications (Murugan & Mishra, 2022). The successful isolation and detailed characterisation of *L. brevis* from *Chubitchi* not only promise it as a native probiotic strain but also demonstrate significant potential for developing region-specific functional starter cultures that enhance both the health benefits and microbial safety of traditional fermented beverages.

3.3.5. Xaj-pani

This traditional alcoholic beverage of the Ahom community in Assam, India, is commonly consumed during religious rites and festivals and enjoyed as a refreshing drink, representing a rich source of gut-modulating microbes (Goswami et al., 2020). It is prepared using a starter culture called *Xaj-pitha*, made from local sticky rice and various medicinal herbs.

Xaj-pitha harbours a diverse microbial consortium including yeasts, moulds, and LAB. The dried cake, *Vekur-pitha*, serves as a repository of *Saccharomyces cerevisiae*, ensuring the continued availability of probiotic yeasts for future fermentation. During fermentation, these microorganisms metabolise carbohydrates and other bioactive compounds, producing ethanol, organic acids, and secondary metabolites that contribute to the characteristic properties of the beverage (Fuloria et al., 2022). The metabolic activity of yeasts and LAB during fermentation generates ethanol, organic acids, and bioactive secondary metabolites that may influence gut microbial composition upon consumption (Fuloria et al., 2022). The microbial profile suggests that regular intake could enrich beneficial gut bacteria, promote microbial diversity, and support the production of metabolites such as SCFAs, which are critical for intestinal barrier function and overall gut homeostasis. These fermentation-derived microbes and metabolites may therefore contribute to the modulation of the GM and the maintenance of gastrointestinal health.

Although *Xaj-pani* may have potential to support immune function, its effects on inflammatory responses and infection-related pathways remain largely unexplored. Further microbiological, metabolic, and clinical studies are required to clarify the health implications of regular consumption and to determine the balance between its probiotic potential and the effects associated with alcoholic fermentation (Fuloria et al., 2022).

3.4. Rice-based fermented foods

3.4.1. Poita Bhat

Rice is a staple dietary source of energy and nutrients for the people of Assam. However, the presence of antinutritional factors, such as phytic acid, limits the bioavailability of essential minerals even after cooking (Ray et al., 2016). The typical form in which this fermented rice dish is consumed is shown in Fig. 6G, illustrating its characteristic soaked and naturally fermented appearance. In a study, the phytate and mineral contents of *Poita bhat* were analysed at different fermentation time points (Goswami et al., 2016). The pH of the fermenting rice decreased from 6.80 to 5.39 within 12 h, reflecting microbial acid production. Concurrently, phytic acid levels declined significantly from 1.255 mg/g in cooked rice to 0.353 mg/g, indicating enhanced mineral bioavailability. Minerals such as magnesium, calcium, iron, and zinc showed increased availability during fermentation, with magnesium being the most abundant and zinc the least. The phytate-to-mineral molar ratios remained well below critical thresholds, further supporting improved nutrient accessibility.

In a study conducted in the Cachar district of Assam, India, samples of fermented sour rice were collected from 23 households (Nath et al., 2021). The traditional preparation method involved soaking cooked rice in water for 12–16 h, after which bacterial isolation and enumeration were performed on MRS agar. The isolates were identified through biochemical tests and 16S rDNA sequencing. *In vitro* assays were performed to assess their probiotic properties, colonisation potential, hemolytic activity, and antagonistic effects against pathogenic bacteria.

The predominant fermentative bacterium was identified as *Weissella confusa* strain GCC_19R1 (Nath et al., 2021).

Beyond nutrient enhancement, fermentation also creates an environment conducive to the growth and activity of beneficial gut microbes. The acidification and partial hydrolysis of antinutritional components during *Poita bhat* fermentation likely promote the proliferation of LAB and other commensal microorganisms, thereby improving gastrointestinal health, enhancing digestion, and modulating the composition of the GM. The identified strain, *W. confusa* GCC_19R1, exhibited robust growth under simulated gastrointestinal conditions, including artificial gastric juice, bile, and pancreatin. Moderate hydrophobicity and autoaggregation indicated its potential for intestinal colonisation. The strain also demonstrated tolerance to acidic pH and 12.5% NaCl and was capable of fermenting glucose, arabinose, and sucrose (Nath et al., 2021). These properties indicate its possible role in maintaining gut microbial balance and supporting digestive health.

Safety assessments confirmed the absence of hemolytic activity in *W. confusa* GCC_19R1, and the strain exhibited antagonistic activity against other bacteria while showing resistance to antibiotics commonly used to treat gastrointestinal infections (Nath et al., 2021). These findings suggest that fermented sour rice harbours microbial strains with probiotic potential that may contribute to gastrointestinal stability, improved nutrient utilisation, and enhanced microbial equilibrium. Overall, the reduction of phytic acid, increased mineral bioavailability, acidification during fermentation, and the presence of probiotic LAB collectively indicate that *Poita bhat* fermentation improves nutritional quality while creating a microbiological environment favorable for gut health (Goswami et al., 2016; Nath et al., 2021).

3.5. Fermented fish and meat-based foods

3.5.1. Hentak

It is a small, ball-shaped traditional fermented fish paste (*Esomus danricus*) from Manipur, which serves as an essential source of protein for the local population (Kabui et al., 2025). LAB were isolated from *Hentak* to identify strains producing antibacterial substances and to optimise their production in the cell-free neutralised supernatant against enteric bacterial pathogens using a “one factor at a time” approach (Aarti et al., 2016).

Among 10 LAB isolates, the most potent strain was identified as *Lactobacillus pentosus*. The strain exhibited potent antibacterial activity in MRS (de Man, Rogosa and Sharpe) medium against indicator enteric bacteria, including *S. epidermidis*, *Micrococcus luteus*, *Shigella flexneri*, *Yersinia enterocolitica*, and *Proteus vulgaris* (Aarti et al., 2017). The cell-free neutralised supernatant was susceptible to pepsin and temperature variations, but remained active over a pH range of 3–7, losing efficacy at higher pH values. The findings from *Hentak* isolates reveal that *L. pentosus* has a dual functional potential, acting as a potent producer of antimicrobial peptides, and *L. brevis* is a resilient probiotic strain with excellent adhesion and tolerance characteristics (Aarti et al., 2017). Together, these strains represent valuable candidates for developing biofunctional starter cultures or probiotic formulations derived from traditional fermented fish products.

3.5.2. Namsing

Namsing is a traditional fermented fish product prepared by the Mising community of Assam from small freshwater fish, and it is commonly consumed as a protein-rich ethnic food in northeastern India (Baruah et al., 2021). The fermentation is carried out naturally, without the addition of defined starter cultures, allowing indigenous microorganisms to drive biochemical and nutritional transformations.

Chakrabarty et al. (2019) investigated the microbial dynamics during *Namsing* fermentation in North Cachar, Assam, by monitoring mesophilic bacteria, lactic acid bacteria (LAB), and Enterobacteriaceae. The mesophilic bacterial population increased rapidly by day 7 and continued to rise gradually until day 28, indicating active microbial

succession throughout fermentation. No yeasts or moulds were detected at any stage, suggesting that the process is predominantly bacteria-driven. LAB were initially absent but later became prominent by day 21, dominating at the final stage, indicating their role in the maturation phase. A total of 27 bacterial isolates with diverse colony morphologies were identified using biochemical and molecular approaches, including *spp.* belonging to *Bacillus*, *Kurthia*, *Providencia*, *Lactobacillus*, *Klebsiella*, *Wohlfahrtiimonas*, *Paenibacillus*, *Staphylococcus*, and *Vagococcus*. The fermentation process, therefore, follows a natural microbial succession dominated by *Bacillus spp.* dominating the early stages and *Lactobacillus spp.* becoming prevalent during maturation. During fermentation, significant biochemical changes were observed, including variations in mineral composition (Chakrabarty et al., 2019). Several isolates exhibited proteolytic activity when tested on casein, indicating their ability to hydrolyse fish proteins into peptides and amino acids that contribute to flavour development and improved digestibility. Among the isolates, *Kurthia gibsonii* showed the highest protease activity, suggesting an important role in protein degradation and sensory enhancement. The progressive microbial succession, together with enzymatic activity, contributes to improved stability, enhanced nutritional quality, and characteristic taste and aroma of the fermented product. These findings also suggest the potential use of selected indigenous strains as starter cultures to standardise fermentation while maintaining traditional characteristics.

Although *Namsing* fermentation is naturally regulated, the presence of diverse bacterial genera, including *Providencia*, *Klebsiella*, and *Staphylococcus*, underscores the need for careful hygienic handling and controlled fermentation conditions to ensure product safety. The absence of moulds and yeasts suggests a relatively stable fermentation environment; however, the use of defined starter cultures and improved processing practices may help minimise contamination risk and enhance microbiological safety in large-scale or commercial production.

3.5.3. *Sathu*

Sathu is a traditional fermented pig fat delicacy widely consumed by the Karbi community of Karbi Anglong, Assam (Chakrabarty et al., 2014). This indigenous product naturally harbours beneficial microorganisms, particularly *Bacillus* and *Lactobacillus spp.* Yukhaibam and Lhouvum (2021) collected *Sathu* from a local Kuki household in Karbi Anglong and isolated bacterial strains to understand its microbial composition and possible health benefits. Two isolates, designated KJC8_C8 and KJR2_C9, were obtained and exhibited distinct morphologies, one being coccoid and the other rod-shaped. Both isolates displayed characteristics consistent with probiotic behaviour, including tolerance to high salt concentration, bile salts, and acidic conditions (Lhouvum, 2021). They also metabolised a wide range of carbohydrate sources, indicating metabolic flexibility. Molecular identification using 16S rDNA sequencing confirmed that these isolates belonged to the *Bacillus* and *Lactobacillus* genera. No spoilage microorganisms were detected in the samples, and the predominance of beneficial bacteria highlighted the microbial safety and potential health-promoting properties of *Sathu*. These observations support the traditional value of *Sathu* and point to its relevance as a naturally enriched microbial food source.

Microbiological analysis of the samples did not detect spoilage microorganisms, indicating a relatively safe fermentation process under traditional preparation conditions. The dominance of beneficial bacteria further supports the product's microbial stability. However, as *Sathu* is produced through spontaneous fermentation, variations in handling, storage, and environmental conditions may influence microbial composition; therefore, adopting controlled fermentation practices and hygienic preparation methods may help ensure consistent quality and safety, particularly if the product is to be standardised or produced at a larger scale.

3.6. Fermented plant-based foods

3.6.1. *Kharoli*

Fermented mustard seed, locally known as *Kharoli* or *Kahudi*, is a distinctive culinary product from Assam, India, prepared using black mustard seeds and the regionally produced alkaline substance called *Khar* (Bora et al., 2022). This traditional preparation is valued not only for its distinctive flavour but also for its potential health benefits, particularly in modulating the GM. The traditional fermentation process involves a complex microbial ecosystem that drives the production of functional metabolites, which may influence gastrointestinal health (Bora et al., 2022).

Structured microbial analysis demonstrated that *Kharoli* samples collected from multiple *Ahom* community households in Moinapara village, Assam, harbour a rich and diverse bacterial community. The dominant genera identified include *Staphylococcus*, *Bacillus*, *Mammaliococcus*, *Weissella*, *Corynebacterium*, *Aerococcus*, *Ruoffia*, *Paenibacillus*, and *Enterococcus* (Das, Tamang, et al., 2024). Multi-omics integration identified the core *spp.* *S. succinus*, *S. saprophyticus*, and *B. subtilis* are correlated with the synthesis of polyunsaturated fatty acids (PUFAs), which are known to influence gut microbial composition and function. Controlled fermentation with this core consortium significantly increased PUFA content to 52.9%, highlighting the potential for targeted microbial interventions to enrich for functional metabolites.

3.6.2. *Panitenga*

Panitenga, another traditional Assamese fermented product prepared using mustard seeds, provides a rich ecological niche for diverse LAB, many of which exhibit promising probiotic and gut-modulatory properties (Sonbhadra & Pandey, 2025). Unlike *Kharoli*, which undergoes alkaline fermentation, *Panitenga* relies on naturally acidic agents, such as *Garcinia Pendunculata*, to create a low-pH environment that supports the selective proliferation of acid-tolerant LAB.

Recent work on *Panitenga* collected from Kamrup district revealed key LAB *spp.* including *Enterococcus durans*, *L. plantarum*, *L. casei* and *L. fermentum* (Sonbhadra & Pandey, 2025). These isolates demonstrated notable physiological and functional traits, including growth under simulated gastrointestinal conditions and biochemical activities such as casein hydrolysis, peptidase and esterase activities, acid phosphatase production, and cell-associated phytase activity. Such enzymatic capabilities not only enhance nutrient bioavailability and reduce antinutritional components but also play a pivotal role in shaping the characteristic sensory profile. The strains showed tolerance to low pH (2-4), bile salts (up to 1%), and salinity (with *E. durans* and *L. plantarum* tolerating up to 6% NaCl), indicating their potential for survival through the gastrointestinal tract. In addition, these LAB isolates exhibited antimicrobial activity against common enteric pathogens, including *S. aureus* and *E. coli*, likely through the combined action of bacteriocins, organic acids, and hydrogen peroxide, as well as the microbial sophistication inherent to this traditional Assamese food system.

3.6.3. *Gundruk*

Gundruk is a traditional fermented vegetable product widely consumed in Sikkim and other Himalayan regions. It is prepared by the natural fermentation of leafy vegetables such as mustard leaves (*Brassica juncea*), radish leaves (*Raphanus sativus*), and cauliflower leaves, which are wilted, crushed, and then allowed to ferment spontaneously under anaerobic conditions (Gautam & Sharma, 2015).

Studies on the microbial diversity of *Gundruk* have identified bacteriocin-producing LAB with potential probiotic relevance. Gautam and Sharma (2015) isolated a novel bacteriocin-producing strain, *Lactobacillus spicheri* G2, using culture-dependent isolation followed by morphological and biochemical characterisation, 16S rRNA gene sequencing, bacteriocin purification, and SDS-PAGE analysis. The strain showed antagonistic activity against several pathogenic and opportunistic bacteria, including *L. monocytogenes*, *S. aureus*, *C. perfringens*, *S.*

mutans, *L. mesenteroides*, and *B. cereus*. The bacteriocin produced by this strain remained stable across a wide pH range and at elevated temperatures, indicating potential stability under gastrointestinal conditions. These observations suggest that the strain may help suppress harmful microbes and favor beneficial bacterial populations; however, the proposed probiotic effects are based mainly on *in vitro* antimicrobial assays. Although the authors suggested that such selective inhibition could contribute to improved gut barrier function, reduced intestinal inflammation, and better microbial balance, these outcomes remain inferential, as no *in vivo* or GM modulation studies were conducted. Therefore, while *L. Spicheri* G2 exhibits promising probiotic-like features; further physiological validation is required to confirm its functional role in gut health and metabolic regulation.

Across the diverse TFFs of Northeast India, several recurring microbial and ecological patterns emerge despite differences in raw materials and preparation methods, thereby enabling a comparative understanding of different fermented products. Fermented soybean products such as *Axone*, *Bekang*, *Hawaijar*, *Kinema*, and *Tungrymbai* are consistently dominated by *Bacillus* spp., reflecting the alkaline fermentation environment that favors spore-forming proteolytic bacteria responsible for protein hydrolysis, peptide formation, and the generation of bioactive metabolites. In contrast, plant-based products prepared from bamboo shoots, vegetables, and cereals, as well as rice-based beverages, typically show a progressive succession of LAB, yeasts, and other acid-tolerant microorganisms, resulting in gradual acidification, detoxification, and improved preservation. Similar microbial transitions are also observed in fermented fish and meat products, where early colonisation by environmental and proteolytic bacteria is followed by the establishment of LAB-dominated communities that stabilise the fermentation matrix and contribute to flavour development and safety. When compared across these different fermentation systems, several functional similarities become evident, including the formation of organic acids, peptides, amino acids, phenolic derivatives, vitamins, and SCFA precursors, which are known to influence gut microbial composition and host metabolic responses. At the same time, each category also exhibits unique characteristics determined by substrate composition, fermentation conditions, and local practices, such as alkaline fermentation in soybean products, acidic fermentation in bamboo shoot and cereal products, and mixed microbial fermentation in fish, meat, and beverage preparations.

A comparative evaluation also reveals common safety-related challenges across these foods. The repeated detection of opportunistic microorganisms, including *Staphylococcus*, *Enterococcus*, *Klebsiella*, *Providencia*, and *Bacillus cereus*, across multiple fermented food types indicates that spontaneous household fermentation may enable the persistence of undesirable microbes when environmental conditions, hygiene practices, or fermentation parameters are poorly controlled. These shared observations demonstrate that, despite the diversity of substrates, TFFs of Northeast India follow broadly similar microbial succession patterns, metabolic transformations, and safety concerns. Such cross-category analysis highlights that the nutritional, functional, and microbiota-modulating properties of these foods arise from common fermentation dynamics rather than from isolated food-specific processes, while still retaining distinctive features shaped by regional ingredients and traditional preparation methods.

4. Nutritional value and pharmacological activities of traditional fermented foods of Northeast India

4.1. Nutritional value

TFF form nutrient-dense components of indigenous diets in Northeast India, where fermentation enhances the bioavailability of vitamins, minerals, amino acids, and other bioactive compounds, thereby preserving nutritional heritage while supporting modern dietary health (Sharma et al., 2025) (Table 2). Bamboo shoot-based foods are

particularly common in Arunachal Pradesh and Manipur, including *Hirring*, *Soibum*, *Soidon*, *Hecche*, *Ekung*, and *Eup*, which are valued for their fibre-rich composition and acidic character that aid digestion (Sonar et al., 2015). These products typically contain high crude fibre, low fat, and moderate protein, with carbohydrates ranging from 7.27 to 25.88%, moisture 31.05–52.58%, and ash 6.22–14.36%, along with phenolics, flavonoids, and tannins that contribute to antioxidant activity (Sonar et al., 2015). *Soibum* provides 362.8 kcal/100 g with proteins, fibre, vitamins, minerals, phytosterols, and polyphenols, while *Soidon* contains 37.2% protein, 46.6% carbohydrates, 5.01% fibre, and low fat, with increased protein and digestibility during fermentation due to microbial proteolysis (Behera & Balaji, 2021; Singhal et al., 2021).

Fermented fish products such as *Ngari*, *Hentaak*, *Shidal*, *Seedal*, *Tungtap*, *Namsing*, and *Napham* are major protein sources across the region. *Ngari* contains 34.1% protein, 33.5% moisture, 21.1% ash, 31.6% carbohydrates, and 13.2% fat, providing 408 kcal/100 g, with minerals including potassium, calcium, sodium, and magnesium (Tamang et al., 2012; Majumdar et al., 2015). *Hentaak* shows high calcium and potassium, along with omega-3 and omega-6 fatty acids, and contains 38.7% carbohydrates, 32.7% protein, and 13.6% fat, with abundant essential and non-essential amino acids. *Puthi shidal* contains up to 38.35% protein, with lysine as the major essential amino acid and palmitic, stearic, oleic, and linoleic acids as dominant fatty acids (Kakati et al., 2018). *Shidal* has higher protein density but lower moisture than *Seedal*, while *Seedal* shows higher pH and lipid stability, both being predominantly bacteria-fermented (Muzaddadi, 2013). *Namsing* retains high protein despite drying, smoking, and fermentation, with only 10.1% moisture, while *Napham* contains 63.65% protein (dry basis) and high mineral content, including sodium, calcium, potassium, zinc, and iron (Baruah et al., 2021; Nath et al., 2024). *Na.kam su.* from the Garo tribe contains 32.54% protein, 36.82% moisture, 16.26% ash, and 13.44% lipid, yielding 419.6 kcal/100 g (Devi et al., 2025).

Soybean-based fermented foods are also nutritionally important. *Kinema* contains 47.7% protein, 17.0% fat, and 28.1% carbohydrates, with *Bacillus subtilis* fermentation improving digestibility and mineral availability (Sarkar et al., 1994). *Tungrymbai* provides 45.9 g protein/100 g dry matter along with fibre, carotene, and folic acid (Agrahar-Murugkar & Subbulakshmi, 2006). *Akhuni* is an alkaline-fermented soybean product (pH ~8.0) rich in digestible amino acids due to *Bacillus*-mediated proteolysis and contains isoflavones such as daidzein, genistein, and glycitein with antioxidant and metabolic benefits (Deb & Bendangnaro, 2020; Das et al., 2025). Other traditional foods also contribute to dietary diversity. *Sa-um*, a fermented pork fat product, contains about 91% lipid with minimal protein and carbohydrate (De Mandal et al., 2018). Fermented vegetables such as *Gundruk* and *Sinki* provide proteins, carbohydrates, minerals, and organic acids, with *Gundruk* containing 38.7% protein and high ash content, while *Sinki* shows lower pH and higher acidity due to lactic fermentation (Senapathi et al., 2016). *Lungsiej* contains 8.5 g protein/100 g dry matter and measurable iron, whereas *Tungtap* provides calcium, phosphorus, sodium, and potassium (Agrahar-Murugkar & Subbulakshmi, 2006). Fermented rice-based foods and beverages also contribute to nutritional intake. *Sai* and *Rohi* show high moisture and active fermentation, with similar sugar and protein content but higher alcohol levels in *Sai*, while *Chasubi*, a rice-flour product, contains 48.08% carbohydrates with moderate protein and low fat, providing 236.6 kcal/100 g (Athokpam & Sidhu, 2024; Kumari et al., 2024).

Overall, TFF of Northeast India provide a wide range of proteins, minerals, fatty acids, vitamins, and bioactive compounds, while fermentation improves digestibility, nutrient availability, and functional properties, highlighting their importance in traditional diets and their potential role in promoting metabolic and gut health.

4.2. Pharmacological activities

TFF have long been recognised for their nutritional and sensory

Table 2
Nutritional composition of some traditional fermented foods from Northeast India.

Local name	State	Tribes	Raw materials	Nutritional composition	References
Fish/Meat Based Fermented Food					
<i>Ngari</i>	Manipur	Meitei	Sundried fish (<i>Puntius sophore</i> Ham.)	Protein: 34.1%; Carbohydrate: 31.6%; Fat: 13.2%; Ash: 21.1%; Moisture: 33.5%Energy: 408 kcal/100 g. Minerals (mg/100g): Ca-362.79, K-58.20, Na-199.66, and Mg-16.06.	Majumdar et al., 2015, Kabui et al., 2025
<i>Hentaak</i>	Manipur	All	Small freshwater fish from the species <i>Esomus danricus</i> (locally known as ngasang) and <i>Alocasia macrorrhiza</i>	Protein: 32.7%; Carbohydrate: 38.7%; Fat: 13.6%; Ash: 15%; Moisture: 40%.	Majumdar et al., 2015, Kabui et al., 2025
<i>Puthi shidal</i>	Tripura	All	Small freshwater fish	Protein: 38.35%; Moisture: 33.44%Lysine (11.8 g/100 g) and glutamic acid (21.4 g/100 g); Fatty acids: palmitic (14.0%), stearic (17.0%), oleic (19.1%), and linoleic (11.0%), indicating cardiovascular and nutritional benefits.	Kakati et al. (2018)
<i>Napham</i>	Assam	Bodo	Smoke dried fish paste and <i>A. macrorrhiza</i>	Protein: 63.65%; Carbohydrate: 6.32%; Fat: 12.12%; Ash: 13.95%; Moisture: 3.52%.Minerals (mg/100g): Na-1064.2, Ca-8,34.6, K-830.2 mg/kg, and essential amino acids (lysine, leucine, valine, isoleucine, methionine)	Nath et al. (2024)
<i>Na.kam su.a</i>	Meghalaya	Garos	Gangetic hairfin anchovy or <i>Setipinna phasa</i> and salt	Protein: 32.54%; Fat: 13.44%; Ash: 16.26%; Moisture: 36.82%; Energy: 419.6 kcal/100 g.	Devi et al. (2025)
<i>Namsing</i>	Assam	Mishing	Fresh water fish, <i>A. macrorrhiza</i> and other local plants	Protein: 54.47%; Moisture: 10.1%; Lipid: 10.57%; Ash: 24.81%	Baruah et al. (2021)
<i>Tungtap</i>	Meghalaya	Khasi	• <i>Puntius</i> and <i>Danio</i> genera fish or fish fat and salt	Protein: 40.6 %; Fat: 19.6%, Ash: 32.2%, Fiber: 0.4%, Minerals (mg/100g): Ca-5040; P-1930 mg/100g; Na-6.26; K-53.18	Murugan & Mishra, 2022
<i>Sa-um</i>	Mizoram	All tribes	• Pork fat	ND	De Mandal et al. (2018)
<i>Kargyaong</i>	Sikkim	<i>Bhutia</i> , Lepcha	• Pork, Beef, Yak meat	ND	Pohnem et al. (2023)
<i>Honoheingrain</i>	Assam	<i>Dimasa</i>	Boar, hog meat•	ND	Pohnem et al. (2023)
<i>Sukuti</i>	Assam	Deori	Harpodon nehereus	ND	Tamang et al. (2016)
Bamboo Based Fermented Foods					
<i>Soibum</i>	Manipur	Meitei	Young shoot of <i>D. hamiltoni</i> , <i>Bambusa manipureana</i> species and <i>Colocasia</i> leaves	Protein: 36.3%; Carbohydrate: 47.2%; Fat: 3.2%; Ash: 13.3%; Fibre: 7.24%; Energy: 362.8 kcal/100 g, phytosterols (3047.30 µg/g), polyphenols (854.7-91,301.10 µg/g), and total flavonoids (408.16 µg/g) Protein: 19.53-27.55%; Carbohydrate: 7.27-25.88%; Fat: 2.19-3.64%; Ash: 6.22-14.36%; Moisture: 31.05-52.58%; Fibre: 3.79-20.84%. Rich in phenolics (718.03-920.10 µg/g), flavonoids (308.72-568.54 µg/g), and tannins (20.09-33.60 µg/g).	Tamang et al. (2012), Kabui et al. (2025)
<i>Hirring</i>	Arunachal Pradesh	Adi, Nyishi	Young fresh shoot of <i>D. hamiltoni</i>	Protein: 37.2%; Carbohydrate: 46.6%; Fat: 3.1%; Ash: 0.96%; Moisture: 92.2%; Fibre: 5.01%; Energy: 363.1 kcal/100 g.	Sonar et al. (2015), Shrivastava et al. (2020)
<i>Soidon</i>	Manipur	Kuki	Succulent <i>Bambusa tulda</i> shoot tip	Moisture: 91.08%; Protein: 3.49%; Fat: 0.05%; Carbohydrate: 1.92%; Ash: 0.80%; Crude fibre: 0.58%; Vitamin C: 0.22 mg/100 g. Minerals (mg/100g): Ca-10.61, Mg-14.91, P-25.38, K-293.24, Na-10.84, Fe-2.89, Zn-0.48, Cu-0.13, Mn-0.10, Se-0.25 mg/100g	Behera and Balaji (2021)
<i>Khorisa</i>	Assam	All	Young shoot of <i>Bambusa balcooa</i>	Protein: 8.5 g/100g; Iron: 1 mg/100g	Singhal et al., 2021, Sharma & Barooah, 2017
<i>Lungsiej</i>	Meghalaya	All	Young bamboo shoot	Protein, amino acids, fatty acids, vitamins, minerals, and carbohydrates The B-vitamin complex included thiamine (B ₁), riboflavin (B ₂), niacin (B ₃), pyridoxine (B ₆), and pantothenic acid (B ₅). The mineral profile comprised potassium (K), manganese (Mn), copper (Cu), calcium (Ca), iron (Fe), and phosphorus (P)	Agrahar-Murugkar et al. (2006)
<i>Kupe</i>	Arunachal Pradesh	Galo	<i>D. hamiltoni</i> , <i>D. hookeri</i> , <i>D. giganteus</i> , <i>D. longispathus</i> , <i>B. balcooa</i> , <i>B. pallida</i> , <i>Melocanna</i>	ND	Tamang et al. (2012), Goyal et al. (2024)
<i>Hikung</i>	Arunachal Pradesh	Nyishi and Galo	Young bamboo shoot of <i>B. tulda</i> , <i>D. hamiltoni</i> , and <i>Phyllostachys bambusoides</i>	ND	Bhardwaj et al. (2024)
<i>Hikung achar</i>	Arunachal Pradesh	All tribes	Fermented bamboo shoots along with <i>Capsicum chinense</i> Jacq. or <i>C. annuum</i> L. or <i>C. frutescens</i> L.	ND	Bhardwaj et al. (2024)
<i>Bastenga</i>	Nagaland	All tribes	<i>D. hamiltoni</i> , <i>B. tulda</i>	ND	Goyal et al. (2024)
<i>Mesu</i>	Sikkim	All tribes	Young shoot of <i>D. hamiltoni</i>	ND	Tamang and Sarkar (1996)
Soybean Based Fermented Foods					
<i>Akhuni</i>	Nagaland	Sumi	Boiled soybean and local plants	Protein: 38.9-42%; Fat: moderate; Carbohydrate: moderate; Fibre: moderate. Alkaline in nature (pH ≈ 8.0)	Deb & Bendangnaro, 2020; Das et al., 2025
<i>Tungrymbai</i>	Meghalaya	Khasi and Garo	Cooked beans and local plants	Protein: 45.9%; Fat: 30.2 g/100g; Fiber: 12.8 g/100g; Carotene: 212.7 µg/100g; Folic acid: 200 µg/100g	Agrahar-Murugkar & Subbulakshmi, (2006)
<i>Kinema</i>	Sikkim	All tribes	Soybeans, firewood ash and local plants	Protein: 47.7%; Fat: 17%; Carbohydrates: 28.1%; Total amino acids: 42,618 mg/100g; Free amino	Sarkar et al. (1994); Senapathi et al. (2016)

(continued on next page)

Table 2 (continued)

Local name	State	Tribe	Raw materials	Nutritional composition	References
				acids: 5129 mg/100g; Ash: 7.2%; Energy: 454 kcal/100g; Minerals (mg/100g): Ca-432; Na-27.7; Fe-17.7; Mn-5.4; Zn-4.5	
<i>Peruyyan</i>	Arunachal Pradesh	Apatani	Cooked soybeans and ginger leaves	ND	Pohnem et al. (2023)
<i>Bekang</i>	Mizoram	All tribes	Dry soybeans	ND	Tamang et al. (2016)
Plant Based Fermented Foods					
<i>Chasubi</i>	Manipur	Meitei	Legumes, cereals	Protein: 10.17%; Carbohydrate: 48.08%; Fat: 0.40%; Fibre: 1.28%; Energy: 236.6 kcal/100 g.	Athokpam and Sidhu (2024)
<i>Gundruk</i>	Sikkim, Arunachal Pradesh	Gorkha, Adi adopted from Gorkha	Dried mustard and raddish leaves	Moisture: 15%; pH: 5.0; Acidity: 0.49%; Ash: 22.2%; Protein: 38.7%; Fat: 2.1%; Carbohydrate: 38.3%; Energy: 321.9 kcal/100 g; Minerals (mg/100g): Ca-234.6; Na-142.2; K-677.6	Senapathi et al. (2016)
<i>Sinki</i>	Sikkim	Gorkha	Tap root of <i>Raphanus sativus</i>	Moisture: 22.8%; pH: 4.1; Acidity: 0.65%; Ash: 15.6%; Protein: 14.9%; Fat: 1.4%; Carbohydrate: 68.0%; Energy: 344.2 kcal/100 g; Minerals (mg/100g): Ca-223.9; Na-737.3; K-2320.4	Senapathi et al. (2016)
<i>Goimoza, Buratamul</i>	Assam, Meghalaya, Mizoram	All tribes	Raw Areca nut, leaflet of areca nut, jute bag, and cow dung	ND	Narzary et al. (2016)
<i>Kharoli</i>	Assam	All tribes	Black mustard seeds and locally made alkali called khar	ND	Narzary et al. (2016)
<i>Panitenga</i>	Assam	All tribes	Black mustard seeds and ripe Thekera (<i>Garcinia pedunculata</i>), and salt	ND	Goswami, Bora, Parveen, Boro, & Barooah (2017)
<i>Anishi</i>	Nagaland	All tribes	<i>Colocasia</i> leaves	Moisture: 6.84%; Ash: 13.22%; Crude Protein: 34.19 g/100g; Crude Fibre: 12.26 g/100g, Energy: 332.18 kcal/100g.	Jamir & Deb, (2018)
Alcoholic Beverages					
<i>Sai Mod</i>	Assam, Arunachal Pradesh	Mishing, Adi,	Rice, roots, leaves and barks of different local herbs	Moisture: 85.5%; Alcohol: 26% v/v; Total soluble sugar: 2.44 µg/mL; Protein: 0.49%; Fat: 0.159 µg/mL	Kumari et al. (2024)
<i>Rohi</i>	Assam	Sonowal, Kachari	Rice, roots, leaves and barks of different local herbs	Moisture: 89.33%; Alcohol: 35% v/v; Total soluble sugar: 2.5 µg/mL; Protein: 0.51%; Fat: 0.154 µg/mL	Kumari et al. (2024)
<i>Tongba</i>	Sikkim	Limbu	brown finger millet (<i>E. coracana</i>) and local herbs	ND	Sen et al. (2025)
<i>Judima</i>	Assam	Dimasa	Bora Rice and bark plant of Thempra (<i>Acacia pennata</i>)	ND	Sarmah et al. (2022)
<i>Zutho</i>	Nagaland	Angami	Sticky rice and banana leaves	ND	Loying et al. (2024)
<i>Duizou/Jou</i>	Nagaland	All tribes	Sticky rice and some local herbs	ND	Loying et al. (2024)
<i>Kiad</i>	Meghalaya	Jaintia and Khasi	Sticky red rice and banana leaves	ND	Loying et al. (2024)
<i>Temi opo</i>	Arunachal Pradesh	Nyishi	Millet and grains of <i>E. coracana</i> L.	ND	Bhardwaj et al. (2024)
<i>Toph opo</i>	Arunachal Pradesh	Nyishi	Maize powder and local rice	ND	Bhardwaj et al. (2024)
<i>Sujen</i>	Assam	Deori	Rice and other local plants	ND	Deka et al. (2025)
<i>Xaj Pani</i>	Assam	Ahom	Rice, roots, leaves and barks of different local herbs	ND	Deka et al. (2025)
<i>Hor-alank</i>	Assam	Karbi	Cooked rice and yeast	ND	Deka et al. (2025)
<i>Atingba</i>	Manipur	Meitei	Local glutinous rice and herbs	ND	Loying et al. (2024)
Milk Based Fermented Food					
<i>Chhu</i>	Sikkim	All tribes	Yak-Milk	Protein: 58.4%; Fat: 5.8%; Carbohydrate: 33.9%; Minerals (mg/100g): Ca-111; Fe-4.5; Mg-64.3; Mn - 3.1.	Dewan & Tamang (2006)
<i>Doi</i>	Assam	All tribes	Raw cow milk	ND	Rai et al. (2016)
<i>Chhurpi</i>	Sikkim, Arunachal	All tribes	Yak milk	Protein: 60-63%, Carbohydrates: 23-24%, Fat: 7-8%, Thiamine: 64.48 µg/g, Vitamin C: 23.53 µg/g, Lactic acid: 0.12-0.23 mg/g Acetic acid: 0.06-0.16 mg/g, Ethanol (trace), Methanol (trace).	Panda et al. (2016)

ND- No Available Data.

attributes, and emerging evidence suggests that it may also confer significant pharmacological benefits (Loying et al., 2024). Traditional fermented products, particularly those derived from plant and soybean sources, are rich in bioactive compounds, including peptides, polyphenols, and isoflavones, which have been shown to modulate metabolic pathways (Loying et al., 2024). Although research remains limited, several studies have reported promising effects of TFF on glucose metabolism and enzyme inhibition, indicating their potential antidiabetic and anti-obesity activity. The integration of such functional foods into the diet may therefore offer a complementary strategy for managing hyperglycemia, obesity and associated metabolic disorders (Wuni et al.,

2022) (Table 3).

Indigenous *Lactobacillus* isolates obtained from *Nakhum* and *Chubitchi* in Meghalaya showed significant pharmacological relevance, exhibiting probiotic and bioactive properties such as α -galactosidase and β -glucosidase activities, high exopolysaccharide production (up to 950 mg/L), and marked proteolytic activity, indicating improved carbohydrate digestion, antioxidant potential, immunomodulatory effects, and bioactive peptide generation (Das et al., 2020). Fermented bamboo shoots of *Bambusa vulgaris* also displayed notable pharmacological activity, with fermentation increasing phenolic and flavonoid content and enhancing antioxidant and anti-inflammatory activity, as confirmed by

Table 3*In vitro* and *in vivo* pharmacological activities of selected fermented foods from Northeast India.

Fermented Product	Region	Model Used	Key Findings	Mechanism/Pathways	Reference
<i>Hawaijar</i> (Fermented Soyabean)	Manipur, India	<i>In vitro</i> (myotubes)	Improved glucose uptake, increased intracellular glucose-6-phosphate	Upregulation of phospho-PI3K, phospho-AKT, phospho-AMPK, GLUT4; bioactive proteins/peptides and isoflavones identified	Das et al. (2023)
		<i>In vivo</i> (HFD-fed rats)	Reduced body weight, fasting blood glucose, HbA1c, insulin resistance, and glucose intolerance	Activation of PI3K/AKT/AMPK/GLUT4 signaling cascade; bioactive proteins/peptides and isoflavones identified	Das et al. (2023)
Fermented rice beer	Assam, India	<i>In vitro</i> (bacterial isolates)	<i>Bacillus</i> sp. FRB_A(A) showed antioxidant activity and α -glucosidase inhibition	PI3K/AKT/AMPK/GLUT4 modulation	Chutia et al. (2025)
		<i>In vivo</i> (STZ-induced diabetic rats)	Reduced fasting blood glucose; restored antioxidant enzymes and glutathione levels	Modulation of oxidative stress; PI3K/AKT/AMPK/GLUT4 pathway	Chutia et al. (2025)
<i>Akhuni</i>	Nagaland, India	<i>In vivo</i> (STZ-induced diabetic mice, 8-week)	Dose-dependent reduction in blood glucose, improved glucose tolerance, decreased HbA1c, increased insulin; mitigated organ pathology	Restoration of pancreatic and hepatic tissue architecture	Das et al. (2025)
<i>Utonga-kupsu</i>	Manipur, India	<i>In vitro</i> microbiological study	Dominant bacterial isolates (<i>B. subtilis</i> , <i>S. nepalensis</i>) exhibited significant antimicrobial, antioxidant, and enzymatic activities.	Production of antimicrobial peptides and antioxidant metabolites, enzymatic degradation releasing bioactive compounds.	Singh et al. (2018)
<i>Hentak</i>	Manipur, India	Experimental microbiological and biochemical characterization	Isolates showed inhibitory activity against <i>S. aureus</i> , <i>E. coli</i> , and <i>P. aeruginosa</i> , indicating their role in food preservation and pathogen suppression	Antibacterial activity through secretion of bacteriocin-like inhibitory substances and enhancement of host defence	Singh et al. (2018)
<i>Ngari</i>	Manipur, India	<i>In vitro</i> probiotic and functional characterization study	Bacterial isolates exhibited high protease, amylase, cellulase, and phosphatase production, improving nutrient bioavailability and gut health.	Modulation of intestinal microbiota, SCFA-mediated gut health improvement.	Singh et al. (2018)
<i>Melye-amiley</i> (Fermented bamboo shoots)	Manipur, India	<i>In vitro</i> (3T3-L1 adipocytes)	Reduced intracellular lipid accumulation	Upregulation of lipolytic genes (HSL, LPL, ATGL) and browning regulators (UCP1, PRDM16, PGC1- α); AMPK activation; mitochondrial biogenesis; fatty acid β -oxidation	Barge et al. (2024)

FRAP, DPPH, and TEAC assays. Major compounds, including protocatechuic acid, p-hydroxybenzoic acid, syringic acid, and kaempferol, contributed to antioxidant, cytoprotective, and anti-inflammatory effects (Singhal et al., 2021).

The antidiabetic potential of *Hawaijar* from Manipur was demonstrated through both *in vitro* and *in vivo* studies (Das et al., 2023). *In vitro* treatment increased glucose uptake and intracellular glucose-6-phosphate in high-glucose-treated myotubes and upregulated phospho-PI3K, phospho-AKT, phospho-AMPK, and GLUT4. *In vivo* administration of *Hawaijar* (50, 100, and 200 mg/kg/day for 16 weeks) in high-fat-diet-fed rats reduced body weight, fasting blood glucose, HbA1c, insulin resistance, and glucose intolerance, while muscle analysis confirmed activation of the PI3K/AKT/AMPK/GLUT4 pathway. SDS-PAGE, immunoblotting, and high-resolution mass spectrometry identified bioactive proteins, peptides, and isoflavones responsible for these effects. Probiotic bacteria isolated from traditionally fermented rice beer of Assam also showed antidiabetic potential (Chutia et al., 2025). *Bacillus* sp. FRB_A(A) and *Acetobacter* spp. FRB_B(S) exhibited antioxidant activity and α -glucosidase inhibition *in vitro*. *In vivo* evaluation of *Bacillus* sp. FRB_A(A) in streptozotocin-induced diabetic rats for 21 days reduced fasting blood glucose from 458.00 ± 46.62 mg/dL to 108.20 ± 6.76 mg/dL, while untreated controls remained hyperglycemic. The treatment also decreased lipid peroxidation and restored antioxidant defence, increasing glutathione, superoxide dismutase, and glutathione peroxidase levels.

Fermented fish products such as *Utonga-kupsu*, *Hentak*, and *Ngari* have also shown pharmacological potential due to their diverse microbial consortia (Singh et al., 2018). Dominant isolates including *Bacillus subtilis* and *Staphylococcus nepalensis* displayed antimicrobial, antioxidant, enzymatic, and probiotic activities, with inhibition of *S. aureus*, *E. coli*, and *Pseudomonas aeruginosa*. DPPH radical-scavenging activity indicated antioxidant peptides, while protease, amylase, cellulase, and phosphatase activities may improve nutrient availability and generate antihypertensive and immunomodulatory peptides. These strains also tolerate gastrointestinal conditions and may modulate GM. Fermented

bamboo shoot extracts were further shown to exert anti-obesity effects in 3T3-L1 adipocytes (Barge et al., 2024), significantly reducing lipid accumulation and upregulating hormone-sensitive lipase, lipoprotein lipase, and adipose triglyceride lipase, along with increased expression of UCP1, PRDM16, and PGC1- α . Activation of AMPK promoted thermogenesis, mitochondrial biogenesis, and fatty acid β -oxidation, highlighting its metabolic benefits. Similarly, the ethanolic extract of *Akhuni* demonstrated antidiabetic activity in streptozotocin-induced diabetic mice (Das et al., 2025). Eight-week treatment produced dose-dependent reductions in blood glucose, improvements in body weight and glucose tolerance, decreases in HbA1c, and increases in insulin levels. Organ-level effects included increased spleen weight and reduced enlargement of pancreas, liver, and kidney, while histological analysis showed restoration of normal pancreatic and hepatic architecture, confirming protective and therapeutic effects.

In addition to the direct pharmacological effects observed in experimental models, the reported activities of these TFFs may also be partly mediated through modulation of the GM. Many of the fermented products discussed above contain LAB, *Bacillus* spp., *Acetobacter* spp., and other beneficial microbes that are known to influence gut microbial composition, enhance SCFA production, and suppress pathogenic bacteria. These microbiota-driven changes can contribute to improved intestinal barrier integrity, reduced endotoxemia, and modulation of inflammatory and metabolic signalling pathways, such as PI3K/AKT, AMPK, and NF- κ B, which are frequently implicated in regulating glucose homeostasis, lipid metabolism, and immune responses. The activation of these pathways observed in studies on *Hawaijar*, fermented bamboo shoots, rice beer probiotics, and fermented fish products may therefore reflect not only the action of isolated bioactive compounds but also the combined effects of fermentation-derived metabolites and interactions with the GM. Nevertheless, in many cases, the mechanistic contribution of microbiota modulation has not been directly investigated, and future studies integrating microbiome analysis with pharmacological evaluation will be essential to clearly establish the microbiota-driven functional effects of TFFs from Northeast India.

5. Safety perspectives

TFF of Northeast India is deeply ingrained in the local food culture because of its nutritional richness, affordability, and long-standing cultural significance (Sithou, 2020). These foods are widely regarded as beneficial for health and are often associated with immune enhancement and protection against seasonal infections. However, most of these products are prepared through spontaneous or back-slopping fermentation under ambient conditions, typically at 28°C–35 °C, usually without refrigeration, preservatives, or strict hygienic control. Such traditional methods, although essential for preserving cultural authenticity, may increase the risk of microbial contamination and variability in product safety and quality (Sithou, 2020). As these foods are produced using diverse raw materials and household-level preparation practices, their microbiological quality can be influenced by several factors, including environmental exposure, handling conditions, fermentation duration, and storage practices, all of which collectively determine the safety of the final product. In addition, the persistence or suppression of undesirable microorganisms during fermentation is influenced by ecological and processing factors, including fermentation temperature, pH changes, salt concentration, oxygen availability, microbial competition, and hygienic practices, which together regulate microbial succession and stability in traditional fermentation. Variations in these parameters may allow opportunistic or toxin-producing microorganisms to survive, particularly in uncontrolled household fermentation systems.

From a microbial hazard perspective, studies have consistently reported the presence of potentially pathogenic microorganisms in TFFs. Large-scale surveillance conducted under the ICMR-FoodNet and MycoNet programs across Assam, Arunachal Pradesh, Tripura, and Sikkim demonstrated that 6.8% of 1227 fermented food samples tested positive for enteric pathogens, with *Bacillus cereus* being the most frequently detected spp. (Das, Tamang, et al., 2024). In addition, other pathogenic bacteria, including *Staphylococcus aureus*, *Escherichia coli*, *Klebsiella pneumoniae*, *Pseudomonas aeruginosa*, *Enterobacter cloacae*, and *Clostridium perfringens*, have been identified in various fermented fish and soybean-based products, particularly under conditions of prolonged fermentation or inadequate hygiene. Such contamination is often attributable to improperly sanitized raw materials, utensils, or water sources. The recurring detection of similar microbial contaminants across diverse fermented food categories suggests that traditional fermentation systems share common microbiological vulnerabilities, especially when carried out under uncontrolled household conditions.

Hygiene-related contamination is further increased during post-fermentation handling and storage, which play a critical role in determining product safety. Several studies have reported coliform counts exceeding acceptable limits, indicating contamination introduced during processing or storage (Albert et al., 2024). The frequent occurrence of *B. cereus* is particularly concerning, as this organism can survive heat treatment and cause gastrointestinal illness when present in high numbers. Beyond direct microbial contamination, antimicrobial resistance (AMR) has emerged as an additional and increasingly important safety issue. LAB isolated from fermented foods, including *Lactobacillus*, *Enterococcus*, *Weissella*, and *Leuconostoc*, have demonstrated multidrug resistance to a wide range of antibiotics such as ampicillin, erythromycin, chloramphenicol, ciprofloxacin, kanamycin, linezolid, streptomycin, trimethoprim, and vancomycin (Albert et al., 2024). Although these organisms are generally regarded as safe, the presence of transferable resistance genes raises concerns regarding their potential horizontal transfer to pathogenic or commensal GM.

In addition to AMR, the formation of biogenic amines has been reported in certain fermented foods (Tamang & Tamang, 2009), posing further toxicological concerns when accumulated beyond safe thresholds. However, such evaluations remain limited for many fermented foods of Northeast India. Similarly, comprehensive probiotic safety assessments, including verification of the absence of transferable

antibiotic resistance genes, virulence-associated traits, and other genomic safety markers, have been conducted in only a small number of studies. This highlights a significant gap in the application of modern safety assessment frameworks to TFFs. Consequently, TFFs that may harbor opportunistic or toxin-producing microorganisms should be consumed with caution, preferably after adequate cooking and proper preparation, particularly when fermentation is performed under uncontrolled household conditions. To systematically integrate these observations, a risk assessment framework is presented (Table 4), incorporating the key components of hazard identification, hazard characterization, exposure assessment, and risk management for TFFs of Northeast India. Owing to the limited availability of population-based epidemiological data, established dose-response relationships, and well-defined consumption patterns specific to these food systems, precise quantification of risk remains constrained. Within this framework, hazard identification and characterization are grounded in available microbiological and toxicological evidence, whereas exposure assessment is derived from reported consumption practices documented in the literature. This approach enables a scientifically informed, relative estimation of risk while acknowledging inherent data limitations.

Taken together, these microbial and biochemical safety concerns challenge the common perception that fermentation inherently ensures food safety. Although fermentation is widely recognised for enhancing the nutritional, functional, and probiotic value of many traditional foods, it should not be universally assumed that all fermented products are inherently safe or beneficial. This complexity is exemplified by fermented areca nut (*Bura tamul*), commonly consumed in Assam. During fermentation, the biochemical profile of the areca nut undergoes significant transformations, including partial reduction of tannins and reduction in alkaloid content such as arecoline (Sun et al., 2024). Epidemiological evidence indicates that the consumption of fermented areca nut, especially when the fermented juice is swallowed, is more associated with precancerous lesions and cancers of the oral and gastrointestinal tract compared to the consumption of fresh areca nut (Patel et al., 2024). This observation reinforces the broader concern that fermentation can, in certain contexts, enhance rather than mitigate health risks. Therefore, fermented foods must be evaluated on a case-by-case basis, as traditional processing methods, microbial ecology, and substrate composition can significantly influence their safety profile. This highlights the importance of incorporating structured safety assessments and risk characterization frameworks when evaluating fermented foods, reinforcing that not all fermented foods confer health benefits and some may pose significant public health concerns.

Despite these concerns, many microbial isolates obtained from TFF also exhibit beneficial properties. Numerous LAB and yeast strains have demonstrated probiotic potential and lack harmful metabolic traits. For example, *L. plantarum* KGL3A, *L. rhamnosus* K4E, *L. fermentum* KGL4, and *Saccharomyces cerevisiae* WBS2A exhibited tolerance to acidic and bile conditions, high cell-surface hydrophobicity, antioxidant activity, cholesterol-lowering ability, and enzymatic activities, including β -glucosidase and α -galactosidase (Ojah et al., 2023). These strains also lacked undesirable characteristics, including hemolysis, urease activity, biogenic amine production, and gelatin hydrolysis, indicating their suitability as starter cultures or probiotic candidates. However, while spontaneous fermentation helps preserve traditional microbial diversity, it also leads to variability and reduced predictability in safety and quality. For this reason, the use of controlled fermentation with selected starter cultures has been suggested as an effective approach to improve product consistency and safety. Studies using starter-based fermentation have shown that fermentation time can be reduced from 4 to 5 months to approximately 75 days, while simultaneously improving microbial safety through pH reduction, increased titratable acidity, suppression of pathogenic *Staphylococcus* spp., and enhancement of beneficial proteolytic and lipolytic activities (Hati et al., 2024). Therefore, improved hygienic handling, careful selection of starter cultures, and routine microbiological evaluation are important to ensure that these foods

Table 4

Safety assessment of the traditional fermented foods of Northeast India.

Hazard Category	Hazard Identification	Hazard Characterization	Exposure Assessment	Risk Management	References
Microbial Pathogens	<i>Bacillus cereus</i> , <i>Staphylococcus aureus</i> , <i>E. coli</i> , <i>Clostridium perfringens</i> , <i>Klebsiella spp.</i>	Enterotoxin production, spore formation (<i>B. cereus</i>), opportunistic infections	Consumption of contaminated soybean, fish, and bamboo-based TFFs; household fermentation	Hygienic practices, safe water, thermal processing, starter cultures	Das, Tamang, et al., 2024; Singh et al., 2014
Hygiene-related Contamination	Coliforms and environmental contaminants	Indicator of fecal contamination and poor sanitation	Post-fermentation handling and storage	GMP, sanitation, proper storage	Albert et al. (2024)
Antimicrobial Resistance (AMR)	Multidrug-resistant LAB (<i>Lactobacillus</i> , <i>Enterococcus</i> , <i>Weissella</i> , <i>Leuconostoc</i>)	Transferable antibiotic resistance genes (ARGs)	Ingestion of live resistant microbes via TFFs	Genomic screening, AMR surveillance, strain selection	Albert et al. (2024)
Opportunistic Microorganisms	<i>Enterococcus</i> , <i>Staphylococcus</i> , <i>Bacillus spp.</i>	Potential virulence traits, toxin production	High microbial load in spontaneous fermentation	Controlled fermentation, proper cooking	Singh et al., 2014; Jeyaram et al., 2008
Biogenic Amines	Histamine, tyramine	Toxic effects at high levels (neurological, cardiovascular)	Formation during prolonged/uncontrolled fermentation	Starter culture selection, fermentation control	Tamang & Tamang, 2009
Fermentation-derived Toxic Compounds	Nitrosamines (e.g., fermented areca nut)	Carcinogenic potential	Chronic consumption, especially fermented juice intake	Process optimisation, nitrite control, awareness	Sun et al., 2024; Patel et al., 2024
Fungal Contamination/ Mycotoxins	Yeasts and moulds	Potential toxin production	Improper storage and moisture conditions	Moisture control, storage regulation	Singh et al. (2014)
Probiotic Safety Concerns	LAB with unknown genetic safety	Presence of ARGs, virulence genes	Direct consumption of live microbes	EFSA/FAO safety evaluation, genomic screening	Borthakur et al., 2022
Lack of Standardization	Variable microbial composition and metabolites	Inconsistent safety and functional outcomes	Widespread spontaneous fermentation	Starter cultures, SOPs, regulatory frameworks	Das, Afzal, et al., 2023; Deka, Handique, & Deka, 2021

retain their traditional characteristics while maintaining acceptable safety standards. A comparative summary of the available evidence for different TFFs, including microbial, functional, pharmacological, genomic, and safety-related studies, is presented in Table 5 to improve clarity and highlight research gaps across food categories.

Overall, the safety of TFFs from Northeast India depends on achieving an appropriate balance between cultural authenticity and microbial control. Strengthening hygienic practices, implementing

regular safety monitoring, and developing region-specific starter cultures can help ensure safe consumption while preserving the unique identity of these ethnic foods. Such an approach can effectively integrate indigenous knowledge with modern scientific understanding, thereby supporting the long-term safety and functional value of these traditional fermented foods (Loying et al., 2024).

Table 5

Evidence level across different categories of traditional fermented foods of Northeast India based on microbial, functional, pharmacological, genomic, and safety studies.

Category	Food	Microbial study	Probiotic	<i>In vivo/In vitro</i> Pharmacological activity	Omics/genomic/ metabolomic	Safety evaluation	References
Soybean	Axone	Yes	Yes	Yes	Yes	Yes	Singh et al., 2014; Das, Afzal, et al., 2023
	Hawaijar	Yes	Yes	Yes	No	Yes	Jeyaram et al., 2008; Das et al., 2023
	Kinema	Yes	Yes	No	Yes	No	Kumar et al. (2019)
	Tungrymbai	Yes	Yes	No	No	No	Chettri & Tamang, 2015
	Bekang	Yes	Yes	No	Yes	Yes	Deka, Handique, & Deka, 2021; Kumar et al., 2023
Bamboo	Akhuni	Yes	Yes	Yes	No	No	Das et al. (2025)
	Khorisa	Yes	Yes	No	Yes	Yes	Sharma & Barooah, 2017
	Soidon/ Soibum	Yes	Yes	No	Yes	Yes	Romi et al. (2015)
	Mesu	Yes	No	No	No	No	Tamang and Sarkar (1996)
	Bastanga	Yes	No	No	No	No	Jamir and Deb (2021)
Beverage	Apong	Yes	Yes	No	Yes	No	Das, Afzal, et al. (2023)
	Atingba	Yes	Yes	No	No	Yes	Mangang et al. (2017)
	Chokot	Yes	No	No	Yes	No	Bhattacharjee et al. (2023)
	Chubitchi	Yes	Yes	No	No	Yes	Murugan & Mishra, 2022
	Xaj-pani	Yes	No	No	No	No	Goswami et al. (2020)
Rice	Poita bhat	Yes	Yes	No	No	Yes	Goswami et al., 2016; Nath et al., 2021
Fish	Hentak	Yes	Yes	No	No	No	Aarti et al. (2016)
	Namsing	Yes	Yes	No	No	Yes	Chakrabarty et al., 2019
	Ngari/Shidal	Yes	No	No	No	Yes	Tamang et al., 2012
Meat	Sathu	Yes	Yes	No	No	No	Yukhaibam and Lhouvum (2021)
	Sa-um	Yes	No	No	No	No	De Mandal et al. (2018)
Plant	Kharoli	Yes	Yes	No	Yes	No	Bora et al. (2022)
	Panitenga	Yes	Yes	No	No	No	Sonbhadra and Pandey (2025)
	Gundruk	Yes	Yes	No	No	No	Gautam and Sharma (2015)

6. Future research opportunities

TFFs of Northeast India represent a rich reservoir of microbial diversity, bioactive metabolites, and functional properties; however, systematic scientific investigation into these foods remains limited. Future research should focus on identifying common microbial patterns and functional similarities across plant, fish, meat, cereal, and beverage-based fermentations to better understand the ecological principles governing traditional fermentation systems. Comparative studies using culture-independent approaches, metagenomics, metabolomics, and functional genomics may help clarify the role of dominant microbial taxa, microbial succession, and metabolite production in determining both safety and health-promoting properties.

One important direction for future work is the development of defined starter cultures derived from indigenous microbial strains. Although spontaneous fermentation contributes to unique sensory and cultural characteristics, it often leads to variability in quality and safety. Isolation, genomic characterization, and functional validation of beneficial strains such as LAB, *Bacillus spp.*, and yeast may allow the design of region-specific starter cultures capable of preserving traditional attributes while improving reproducibility, shelf life, and microbiological safety. In this context, whole-genome sequencing, resistome analysis, and evaluation of virulence factors, toxin genes, and biogenic amine production should be incorporated as standard criteria for the selection of probiotics and starter cultures.

The application of systems biology and multi-omics approaches will also be essential for understanding the complex interactions among microorganisms, substrates, and environmental factors during fermentation. Integration of metagenomics, transcriptomics, proteomics, and metabolomics can provide deeper insight into metabolic pathways responsible for the production of SCFA, bacteriocins, exopolysaccharides, vitamins, and other bioactive compounds that contribute to GM modulation and host health. Such approaches may also help identify key ecological parameters, including pH, salt concentration, temperature, and microbial competition, that influence pathogen suppression and fermentation stability. Another important research priority is the standardisation of traditional fermentation processes without compromising cultural authenticity. Establishing controlled fermentation protocols, hygienic handling practices, and quality control measures can reduce the risk of contamination, dissemination of antimicrobial resistance, and toxin formation. At the same time, documenting indigenous knowledge and traditional preparation methods is necessary to ensure that scientific standardization does not lead to the loss of cultural heritage. In addition, although recent studies have begun to address important safety parameters, such as biogenic amine formation. We note that studies on antimicrobial resistance and comprehensive genomic-level evaluations remain limited, resistome analysis and genome-based safety assessment have been performed only for a few isolates, and such investigations remain scarce for many TFFs in this region. Therefore, further studies incorporating whole-genome sequencing, functional gene annotation, and detailed safety profiling are required to fully establish their safety in accordance with current probiotic and fermented food safety guidelines.

A key safety concern identified across the studies is the absence of quantitative microbial risk assessment, as most reports focus only on detecting microorganisms without assessing whether their levels exceed acceptable safety limits. This limitation restricts the proper evaluation of potential public health risks associated with consumption. Therefore, future investigations should prioritize quantifying pathogen loads and establishing their relevance to food safety standards to ensure consumer protection. From a translational perspective, greater attention should be given to the industrial application and commercialization potential of TFFs of Northeast India. Scientific validation of safety, nutritional value, and pharmacological activity may facilitate the development of functional foods, probiotics, and nutraceutical products. However, successful commercialization will require compliance with modern food safety

regulations, including microbial risk assessment, genomic safety evaluation, and standardisation of production methods. Future research should prioritize strain-level identification of key fermenting microorganisms, alongside integrative multi-omics approaches, including metabolomic and microbiome-based characterization of bioactive components, to better elucidate their functional roles. Only a single genomic-level investigation has been reported to date (Huidom et al., 2024), highlighting a significant gap in the comprehensive molecular characterization of TFFs of Northeast India. Moreover, most existing studies have relied on metagenomic analyses, which, while informative about microbial diversity and community structure, provide limited insight into functional activity and biochemical outputs. In particular, there remains a marked paucity of proteomic and metabolomic investigations capable of elucidating specific metabolic pathways, enzymatic interactions, and bioactive metabolite profiles that underpin the health-promoting properties of these foods. Bridging these gaps through integrated, multi-omics approaches would enable a deeper mechanistic understanding of how TFFs modulate GM and influence metabolic regulation. Furthermore, rigorously designed preclinical studies and well-controlled clinical trials are essential to substantiate the therapeutic potential of these foods in the management of metabolic disorders. In parallel, the development of standardized starter cultures, implementation of quality-control measures, and formulation of functional foods or nutraceuticals derived from selected TFFs will be crucial to ensure safety, reproducibility, and scalability. Importantly, there is also a pressing need for more comprehensive quantitative risk characterization, supported by population-based data, dose-response relationships, and detailed consumption patterns, to enable more accurate evaluation of potential health risks and to strengthen the scientific basis for safe commercialization and regulatory standardization. Such a comprehensive strategy will support the effective integration of traditional knowledge with modern nutritional science, while safeguarding cultural heritage, ensuring product safety, and promoting sustainable food innovation in Northeast India.

7. Conclusion

TFFs of Northeast India reflect a unique interplay of biodiversity, cultural heritage, and microbial richness, unlike most other regions of India, where dairy-based fermented products such as curd, lassi, and buttermilk predominate. A diverse range of plants, rice, fish, and meat-based fermented foods characterise Northeast India. Ecological abundance shapes this distinct fermentation tradition, access to forest and freshwater resources, minimal historical emphasis on large-scale dairy systems, and long-standing ethno-culinary practices among diverse tribal communities, who relied on fermentation as a natural method of food preservation and flavour development. These TFFs harbour beneficial microorganisms, including *Lactiplantibacillus*, *Limosilactobacillus*, *Lactocaseibacillus*, *Pediococcus*, *Enterococcus*, and spore-forming *Bacillus* and *Paenibacillus spp.*, which produce organic acids, exopolysaccharides, vitamins, peptides, phenolic derivatives, and other bioactive compounds. Emerging evidence indicates that several isolates from these foods can modulate key metabolic pathways, including PI3K/AKT, AMPK, and GLUT4 signalling, suggesting promising implications for gut health, immune modulation and metabolic regulation. However, despite the isolation of numerous beneficial strains, only a limited number have been systematically studied for GM modulation and host responses, or have been validated through *in vivo* and mechanistic studies, highlighting a critical research gap.

One of the central themes of this review is also the need for scientific validation, standardisation and safety assurance. Most TFF in the region are prepared at the household level through spontaneous fermentation, which may lead to inconsistent microbial composition and potential risks of pathogenic contamination, including the incidental presence of *Staphylococcus spp.*, *Enterobacteriaceae*, *Clostridium spp.*, and *B. cereus*, as reported in some studies. Such variations can also contribute to the

accumulation of biogenic amines and the circulation of AMR. Strengthening hygienic practices, introducing routine microbial safety assessments, and developing affordable, region-specific starter culture systems are essential to ensure safe, reproducible fermentation while preserving traditional knowledge. Also, the fermented foods discussed in this review are limited to those currently available in the scientific literature and therefore do not represent a complete inventory of TFFs of Northeast India. Many traditional preparations remain confined to tribal households and remote communities, where they are practiced without formal documentation or scientific investigation. In addition, the available literature is uneven across food types, geographical regions, and ethnic groups, with certain well-known products being repeatedly studied while many others remain underreported or unexplored. This imbalance may partly reflect differences in research accessibility, commercial interest, and methodological feasibility, which can introduce selection bias in the current understanding of microbial diversity and functional properties of these foods. Further ethnographic surveys and detailed microbiological studies of these lesser-known fermented foods may reveal additional microbial diversity and uncover novel functional and therapeutic properties. Integrating traditional wisdom with modern microbial science and regulatory frameworks can facilitate the development of culturally grounded functional foods, next-generation probiotics, and nutraceuticals from this region.

Overall, this review provides a more comprehensive and integrative perspective by systematically compiling evidence on microbial diversity, metabolite composition, functional health relevance, safety, and the translational potential of these foods. Special emphasis is placed on their possible roles in GM modulation, metabolic health regulation, and the development of microbiome-targeted functional foods, which have not been collectively addressed in earlier reports. By linking traditional fermentation practices with contemporary findings in nutrition, microbiology, and pharmacology, this review establishes a broader scientific framework that supports the functional and therapeutic relevance of these indigenous foods and highlights their potential for future application in evidence-based nutrition and health research.

Despite promising *in vitro* and *in vivo* studies, clinical evidence supporting the health benefits of these TFFs remains limited. Future research should focus on well-designed clinical trials to validate their safety and therapeutic efficacy in humans. In addition, the doses used in animal studies may not accurately reflect typical human dietary intake; therefore, the inclusion of human-equivalent dose considerations is necessary to improve the translational relevance of these findings. Furthermore, chronic toxicity and long-term safety evaluations of these fermented foods are largely lacking, representing an important research gap that should be addressed in future studies to ensure their safe and effective use. With well-defined safety evaluation, standardized fermentation protocols, and systematic translational studies, the TFFs of Northeast India can be advanced from ethnically preserved dietary practices to scientifically validated functional foods. Future research should prioritize strain-level identification of key fermenting microorganisms, metabolomic and microbiome-based characterization of bioactive components, and controlled preclinical and well-designed clinical studies to confirm their role in GM modulation and metabolic disorder management. In parallel, the development of standardized starter cultures, quality-control guidelines, and functional-food or nutraceutical formulations based on selected TFFs will be essential for safe commercialization. Such an approach will enable the integration of traditional knowledge with modern nutritional science while ensuring cultural preservation, product safety, and sustainable food innovation in Northeast India.

CRedit authorship contribution statement

Lunasmrita Saikia: Writing – original draft, Data curation, Conceptualization. **Partha Pratim Dutta:** Writing – review & editing, Supervision, Formal analysis, Data curation, Conceptualization.

Dhrubajyoti Gogoi: Writing – original draft, Data curation. **Narayan Chandra Talukdar:** Visualization. **Saikat Sen:** Supervision. **Prastuti Sahariah:** Writing – original draft. **Albert Bharali:** Writing – original draft.

Declaration on the use of AI tools

The author(s) declare that AI tools, specifically ChatGPT by OpenAI, were used solely to paraphrase some sentences and substitute certain words to improve language and reduce potential plagiarism. No content, including original sentences, figures, or images, was generated using AI tools. All scientific content and interpretations are entirely the work of the author(s).

Funding

No funding available.

Declaration of competing interest

The authors state there is no conflict of interest.

Acknowledgement

We gratefully acknowledge Assam down town University, Assam, India, for providing the necessary research facilities and financial support that enabled the successful completion of this work. We also extend our sincere thanks to Noksensangla Y. Chang, Aniya Teli, Ruth Laltlannungi, Pallab Deori, Shivam Rongpi, Binyom Kamduk and Benjamin Das for providing photographs of the fermented food used in this manuscript. The illustrations and graphical representations were prepared using BioRender (www.biorender.com).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.fbio.2026.108992>.

Data availability

No data was used for the research described in the article.

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